

**PHYSICS-INFORMED RESIDUAL LEARNING
ARCHITECTURE FOR VIRTUAL FLOW
METERING IN THE OIL AND GAS INDUSTRY**

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Master's in Data Science and Artificial Intelligence

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Faculty of Engineering

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DECLARATION

I declare that this is my own work and this Dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. I retain the right to use this content in whole or part in future works (such as articles or books).

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The supervisor should certify the Dissertation with the following declaration.

The above candidate has carried out research for the Master's in Data Science and Artificial Intelligence Dissertation under my supervision. I confirm that the declaration made above by the student is true and correct.

Name of Supervisor: Dr. Uthayasanker Thayasivam

Signature of the Supervisor:

Date:

DEDICATION

To my beloved wife and parents, your unwavering love, encouragement, and sacrifices have been the cornerstone of my academic journey. Your constant support gave me the strength to pursue my dreams and overcome challenges along the way.

ACKNOWLEDGEMENT

I extend my sincere appreciation to Dr. Uthayasanker Thayasivam from the University of Moratuwa for the invaluable supervision and insightful guidance throughout the course of my research.

ABSTRACT

Accurate estimation of multiphase flow rates is essential for effective production monitoring and reservoir management in the oil and gas industry. Although Multiphase Flow Meters provide direct measurements of oil, gas, and water rates, their high cost and operational complexity limit widespread deployment, leading to sparse and irregular flow rate observations. Virtual Flow Metering offers a software-based alternative; however, conventional physics-based VFM models are often constrained by simplifying assumptions and parameter uncertainty, while purely data-driven approaches require large volumes of labeled data and may produce non-physical predictions.

This thesis proposes a physics-informed residual learning architecture for virtual flow metering that integrates first-principles physical modeling with data-driven machine learning. A baseline physics-based flow model is first calibrated independently for each well using routinely available pressure, temperature, and choke measurements. A global machine learning model is then trained to learn the residual errors between physics-based predictions and measured flow rates, thereby capturing unmodeled multiphase flow effects and systematic model discrepancies. During inference, physics predictions and learned residual corrections are combined through a hybrid mechanism governed by physically motivated gating rules to ensure robustness and physical consistency.

The proposed framework is evaluated using historical multi-well production data under sparse measurement conditions. Model performance is assessed using standard regression metrics and compared against physics-only and purely data-driven baseline approaches. Results demonstrate that the physics-informed residual learning framework improves flow rate estimation accuracy while maintaining interpretability and robustness, highlighting its suitability for practical deployment in industrial virtual flow metering applications.

Keywords: Virtual Flow Meter, Oil and Gas Industry, Physics Informed Machine Learning

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CHAPTER 1

INTRODUCTION

The oil and gas industry is a global sector focused on the exploration, extraction, production, transportation, refining, and commercialization of hydrocarbons. As one of the most critical energy industries, it supplies fuels, feedstocks, and raw materials essential for industrial, commercial, and domestic use. The industry is characterized by complex operations spanning multiple technical, economic, and regulatory domains, requiring integration of geology, engineering, logistics, and environmental management. Hydrocarbon resources are typically categorized into crude oil, natural gas, and natural gas liquids, each of which undergoes specific extraction, processing, and distribution workflows. The industry operates across a value chain divided into three principal segments: Upstream, which involves exploration and production; Midstream, which focuses on transportation, storage, and initial processing; and Downstream, which encompasses refining, distribution, and commercialization of end-use products (Lake, 2006; Ahmed, 2010).

The Upstream sector encompasses all activities related to the exploration and production of hydrocarbons. This includes geological and geophysical surveys, exploration drilling, reservoir characterization, well planning, drilling operations, completions, and production of oil, gas, and condensate. It is the segment where subsurface engineering, reservoir management, and field development strategies are applied. Central to these operations are fields and reservoirs, which define the spatial and geological context for upstream activities. A field is a geographically or geologically defined area that contains one or more hydrocarbon reservoirs. Fields represent the primary operational units for planning drilling programs, designing production facilities, and managing overall hydrocarbon recovery. They are delineated based on subsurface geology, reservoir continuity, hydrocarbon presence, and operational or administrative considerations.

An oil well is a borehole drilled from the surface into a hydrocarbon-bearing reservoir to access and produce oil, gas, and associated fluids. Wells are designed to provide a controlled pathway for hydrocarbons to flow from the reservoir to the surface, and their design depends on reservoir properties, depth, pressure, and fluid composition. The fluid mixture flowing out of the well consists of oil, natural gas and water. This mixture of flow is called the multi-phase flow. The flow rates of oil, gas, and water reflect the performance of the well. The Petroleum Engineers need to make informed decisions regarding the production strategies based on the performance of the wells. Through accurate predictions of flow rates, petroleum Engineers can diagnose production inefficiencies and constraints, and implement solutions to optimize production and project future investments. Physical flow meters and Virtual Flow Meter (VFM) are

used to predict flow rates in oil wells. Physical flow meters are often very expensive, require regular maintenance, and are prone to failure under harsh operating conditions. Physical flow meters use in-built sensor data and physics based mathematical equations to predict the real-time flow rates. Virtual flow meters, in contrast, can use mathematical models based on physics, models based on machine learning, or hybrid models to predict flow rates ([Bikmukhametov and Jäschke, 2020b](#)).

1.1 Asset Hierarchy

In upstream oil and gas operations, the management and organization of hydrocarbon resources are structured through a hierarchical framework known as the asset hierarchy [1.1](#), which provides clarity for both subsurface and surface operational planning.

- **Asset:** An Asset is the highest-level operational and resource unit in upstream oil and gas management, encompassing multiple geographically or geologically defined oil and gas fields. It serves as a consolidated management entity under which exploration, development, production, and economic evaluation of the associated fields are coordinated. The asset framework enables integrated planning of drilling campaigns, reservoir development, production optimization, and infrastructure utilization across all fields it contains. By grouping multiple fields under a single asset, operators can streamline decision-making, allocate resources efficiently, and evaluate overall performance, reserves, and recovery potential at a portfolio level. In essence, the asset represents the top-level resource that aligns subsurface resources, surface facilities, and economic objectives for strategic management.
- **Field:** A geographically bounded development area within the asset that contains surface facilities and one or more subsurface hydrocarbon-bearing reservoirs. The field is the primary operational unit for drilling, production, and facility management.
- **Reservoir:** A distinct geological formation within the field that contains hydrocarbons in porous and permeable rock. Reservoirs may be stacked vertically or distributed laterally. They define the subsurface domain for fluid flow and reservoir engineering analysis.
- **Sector:** A subdivision of a reservoir or field created for reservoir simulation, surveillance, or operational segmentation. Sectors capture heterogeneity, fault compartments, pressure regimes, or modeling boundaries that enhance analytical accuracy.

- **Cluster:** A group of wells or subsurface drainage areas within a sector or field that are managed together based on operational similarity, shared production routing, or common surveillance requirements. Clusters serve as practical organizational units for production monitoring, optimization, and VFM-based diagnostics.
- **Wellpad:** A Wellpad is a surface facility that hosts one or more wells drilled into the subsurface to access hydrocarbon reservoirs. Wellpads provide the interface between subsurface production and surface infrastructure, including drilling rigs, flowlines, manifolds, and gathering systems. They are strategically located to optimize drilling efficiency, minimize environmental footprint, and facilitate centralized management of multiple wells. Wellpads serve as the operational unit for drilling campaigns, completions, and surface-based production monitoring, and they often group wells that belong to the same operational cluster or reservoir drainage area.
- **Well:** A Well is an individual borehole drilled from a wellpad into a reservoir to extract hydrocarbons or inject fluids. Each well represents a discrete production or injection point and can be monitored, managed, and optimized independently. Wells are assigned to clusters based on shared production characteristics or operational strategies, enabling efficient resource management. They may be vertical, deviated, or horizontal depending on reservoir geometry and development objectives. Wells serve as the primary conduit for connecting subsurface reservoirs to surface production systems.
- **Wellbore / Well String:** A Wellbore, often referred to as a well string, represents the drilled path of a well, including all installed casing, tubing, completions, and downhole equipment. A single well may contain multiple wellbore strings targeting different zones, formations, or reservoirs, allowing for separate production streams, reservoir compartment management, or enhanced recovery strategies. Wellbore design directly influences flow capacity, pressure profiles, and production efficiency, and it provides the physical framework for subsurface monitoring, artificial lift installation, and advanced reservoir management techniques.

1.2 Oil and Gas Wells

Hydrocarbon production wells are broadly classified into oil wells and gas wells based on the dominant produced phase. In practice, both types involve multiphase flow consisting of oil, gas, and water, which introduces significant complexity in modeling and

measurement. Accurate estimation of phase flow rates is essential for reservoir management, production optimization, and allocation (Falcone et al., 2009; Kolev, 2016).

1.2.1 Oil Wells

Oil wells primarily produce liquid hydrocarbons, often accompanied by associated gas and water. Their behavior is typically governed by multiphase liquid-dominated flow.

1.2.1.1 Characteristics

- Oil is the dominant phase, with gas present as associated or dissolved gas.
- Water production increases over time due to reservoir depletion and water breakthrough.
- Flow is multiphase and generally liquid-dominated.
- Artificial lift methods such as gas lift or Electric Submersible Pumps (ESP) are often required.

1.2.1.2 Flow Behavior

- Flow is governed by pressure drawdown and inflow performance relationships.
- Multiphase interactions between oil, gas, and water lead to complex flow regimes (Beggs and Brill, 1994).
- Water cut (wc) is a key parameter describing liquid composition:

$$wc = \frac{q_w}{q_L}, \quad q_L = q_o + q_w$$

- Increasing water cut impacts production efficiency and measurement accuracy (Falcone et al., 2009).

1.2.1.3 Measurement Challenges

- Accurate phase separation is difficult, particularly at high water cut.
- Multiphase Flow Meters (MPFM) exhibit varying accuracy depending on flow regime (Falcone et al., 2009).
- Well test data is reliable but sparse, whereas continuous sensor data is indirect and requires modeling.

1.2.2 Gas Wells

Gas wells primarily produce natural gas, with relatively small amounts of condensate and water. Their behavior is typically gas-dominated and strongly influenced by pressure conditions.

1.2.2.1 Characteristics

- Gas is the dominant phase, often produced at high velocities.
- Liquid phases are typically minor but can significantly affect flow behavior.
- Flow rates are highly sensitive to pressure differences between reservoir and wellbore.

1.2.2.2 Flow Behavior

- Gas flow is commonly modeled as a pressure-driven process:

$$q_g \propto \sqrt{P_{res} - P_{wf}}$$

- Strong dependence on reservoir pressure and surface conditions.
- Liquid loading can occur when gas velocity is insufficient to transport liquids, reducing production efficiency ([Beggs and Brill, 1994](#)).

1.2.2.3 Measurement Challenges

- Gas flow measurements are sensitive to pressure and temperature variations.
- Small amounts of liquid can significantly impact measurement accuracy.
- Indirect measurement techniques such as virtual flow metering are often required for continuous estimation.

1.2.3 Relevance to Virtual Flow Metering

Multiphase flow measurement remains a major challenge due to the simultaneous presence of oil, gas, and water phases. Traditional approaches such as test separators provide accurate but infrequent measurements, motivating the development of virtual flow metering (VFM) techniques ([Falcone et al., 2009](#)).

Hybrid approaches that combine physics-based modeling with data-driven techniques have shown improved performance in handling complex multiphase systems, as they leverage both physical understanding and data-driven adaptability ([Kolev, 2016](#)).

1.3 Producer and Injector Wells

- **Production Wells:** Production wells are the primary pathways through which hydrocarbons - oil, gas, and associated formation water, are extracted from a reservoir and transported to surface processing facilities. Their core function is to withdraw fluids from subsurface formations in a controlled manner that optimizes recovery while maintaining well integrity and reservoir performance. The performance of a production well is driven by the pressure differential between the reservoir and the wellbore (Ahmed, 2010). Early in a field's life, natural energy such as solution gas drive, water drive, gas cap expansion, or aquifer support pushes fluids toward the wellbore. As reservoir pressure declines over time, production wells may transition to artificial lift systems such as gas lift, Electric Submersible Pumps (ESP) , Progressive Cavity Pumps (PCP) , or Rod Pumps (RP) to sustain flow rates and overcome the hydrostatic and frictional losses in the wellbore (Brown, 1984). The choice of artificial lift depends on fluid properties, well depth, deviation, GOR, water cut, and economic considerations.
- **Injection Wells:** Injection wells are specialized wellbores designed to introduce fluids - such as water, gas, or chemical mixtures into subsurface formations for the purposes of reservoir pressure maintenance, Enhanced Oil Recovery (EOR) (Lake, 2014), or safe disposal of produced water. Unlike production wells, which are engineered to withdraw hydrocarbons, injection wells primarily function as input conduits, pushing fluids into the reservoir to support long-term field productivity. Their role is essential in secondary and tertiary recovery processes, as they improve sweep efficiency and maintain reservoir pressure, thereby enhancing the deliverability of adjacent production wells.

1.4 Well Downhole and Surface Instrumentation

The configuration of instrumentation in an oil well varies depending on the operational objectives and reservoir characteristics. In general, sensors are strategically installed along the wellbore and surface facilities to monitor drilling, production, and safety-related parameters. The wellbore, often referred to as the borehole, is the cylindrical hole drilled from the surface into subsurface formations during the development of oil, gas, water, or geothermal wells. Figure 1.2 illustrates the typical placement of instrumentation across different locations of a producing well.

1.4.1 Downhole Instrumentation

Well Down Hole (WDH) instrumentation refers to sensors installed at or near the bottom of the well, often integrated within downhole tools or permanent monitoring systems. These sensors are designed to capture critical information about the subsurface environment and wellbore conditions. Commonly measured parameters include pressure, temperature, and fluid properties, with some advanced systems also capable of acquiring seismic or acoustic data.

The primary downhole instruments used in this study are the Down Hole Pressure (DHP) and Down Hole Temperature (DHT) sensors. These measurements provide real-time insight into reservoir behavior and well performance, enabling informed decision-making related to production optimization, reservoir management, and early detection of anomalies. Downhole data are particularly valuable for assessing reservoir drawdown, diagnosing flow restrictions, and ensuring operational safety throughout the well lifecycle.

1.4.2 Wellhead Instrumentation

Well Head (WH) sensors are installed at the wellhead, which represents the interface between the subsurface wellbore and surface production facilities. Wellhead instrumentation plays a crucial role in monitoring surface-operating conditions and ensuring safe and efficient production.

Typical wellhead measurements include pressure and temperature, obtained using the Well Head Pressure (WHP) and Well Head Temperature (WHT) sensors, respectively. These measurements are essential for monitoring well integrity, detecting abnormal operating conditions, and supporting production control strategies. Wellhead data also serve as key inputs for production surveillance systems and virtual flow metering applications.

1.4.3 Downstream Choke Instrumentation

The choke system is located downstream of the wellhead and is used to regulate the flow of produced fluids from the well. Choke valves control the flow rate by introducing a pressure drop, thereby enabling operators to manage reservoir drawdown, prevent sand production, and maintain well integrity.

Instrumentation associated with the choke includes the Downstream Choke Valve (DCV), which represents the choke opening or valve position, and the Downstream Choke Pressure (DCP), which measures the pressure downstream of the choke. Choke valves may be manually operated or automatically controlled as part of a production control system. Their placement downstream of the wellhead provides flexibility in regulating flow while maintaining safe operating conditions.

1.4.4 Role of Instrumentation in Virtual Flow Metering

All or a subset of the aforementioned sensors are used as independent variables in machine learning (ML)-based Virtual Flow Metering (VFM) models. Measurements from downhole, wellhead, and choke locations collectively capture the pressure–temperature–flow relationships governing multiphase production. The integration of these sensor signals enables data-driven models to infer oil, gas, and water flow rates in the absence of direct physical flow meters, thereby supporting real-time production monitoring and optimization.

Table 1.1
Well downhole and surface instrumentation

Sensor	Description	Unit
dhp	downhole pressure	bara
dht	downhole temperature	°C
whp	wellhead pressure	bara
wht	wellhead temperature	°C
dcp	Downstream choke pressure	bara
choke	Downstream choke regulation	%

1.5 Flow States

Fluid flow within an oil and gas well occurs through a complex piping network and exhibits a wide range of physical behaviors. The flow is inherently multiphase and is influenced by well geometry, flow direction, and temporal variations in operating conditions. These characteristics collectively contribute to the complexity of flow-rate measurement and modeling in producing wells. Fluid flow in oil and gas wells typically consists of a multiphase mixture of oil, gas, and water. The relative proportions of these phases vary spatially along the wellbore and temporally with changing reservoir and operating conditions. Multiphase interactions give rise to complex flow phenomena such as phase slip, interfacial instabilities, and regime transitions, which significantly complicate both physical measurement and data-driven estimation of flow rates (Beggs and Brill, 1994). The piping network of an oil and gas well may include vertical, deviated, inclined, and horizontal sections. As a result, fluid flow can occur in upward, downward, or mixed directions depending on the well trajectory. Changes in inclination strongly affect phase distribution, gravitational segregation, and pressure losses, thereby influencing the observed flow behavior and sensor measurements along the well.

1.5.1 Steady State Flow

Steady state flow refers to operating conditions under which the pressure, flow rates, and phase fractions of the produced fluids remain approximately constant over time. In this regime, spatial and temporal gradients of flow variables are relatively small, and the system can be reasonably assumed to be in equilibrium. Steady-state assumptions are commonly employed in analytical flow models and production surveillance due to their mathematical simplicity and reduced computational complexity. However, such assumptions may not fully capture the dynamic behavior of multiphase flow in real well operations ([Shippen and Bailey, 2012](#)).

1.5.2 Unsteady State Flow

Unsteady state flow is characterized by temporal variations in pressure, flow rates, and phase distribution within the wellbore. These fluctuations may arise from operational changes such as choke adjustments, variations in reservoir inflow, or transient flow phenomena including slugging and phase redistribution. In unsteady conditions, the flow variables are continuously evolving, violating steady-state assumptions and increasing the uncertainty of flow-rate measurements. Capturing unsteady-state behavior is therefore critical for accurate production monitoring and motivates the use of advanced data-driven and hybrid modeling approaches ([Santim et al., 2020](#)).

1.5.3 Implications for Flow Measurement

The combined effects of multiphase flow, complex well geometry, variable flow direction, and unsteady operating conditions present significant challenges for accurate flow-rate measurement in oil and gas wells. These complexities limit the reliability of conventional physical flow meters and motivate the use of virtual flow metering approaches for production monitoring ([Bikmukhametov and Jäschke, 2020b](#)).

1.6 Physical Flow Meters

Physical flow meters encompass a diverse range of instruments designed to measure the rate of fluid flow across various industries and applications. These meters operate based on physics principles, employing mechanisms like pressure differentials, velocity measurement, displacement, mass calculation, or volumetric methods. Their versatility lies in their ability to cater to a wide array of fluids, from gases and liquids to multi-phase mixtures. Even in the physical flow meters, flow rates are calculated indirectly using physics based mathematical equations since it's impossible to have flow rates measured directly. Physical flow meters in general have the limitation of not being able to back-fill historical flow rates prior to the installation of the instrument.

1.6.1 Single Phase Flow Meter

Traditional Single Phase Flow Meter (SPFM) systems necessitate the complete separation of the phases within the well streams before the measurement. In production metering, this separation typically occurs naturally at the exit of a conventional processing facility. These plants are designed to amalgamate various well streams at one end and deliver stabilized individual phases for transport at the other end. Typically, single-phase metering systems offer top-notch accuracy in measuring hydrocarbon production ([Corneliussen et al., 2005](#)). In production system a Well Test (WT) separator is used to physically separate the individual phases of the produced fluid so that each phase can be accurately measured and analyzed. Once the phases are separated, dedicated SPFM are employed to measure the flow rates of each individual phase of oil, gas and water. Although not as advanced as multi-phase counter parts, the SPFM are also considered expensive and its usage is extremely costly due to production deferment. This is because the well has to be disconnected from the production and needs to be connected with the separator prior to feeding the SPFM. Single phase flow measurement is often called a well test. It's possible that some wells cannot be measured from a well test because the distance from the well to the separator can be very long. Oilfields are required to perform well-tests at some point based on regulatory requirements of the jurisdictions. It is however not economically feasible to run regular well-tests as they cause production deferment.

1.6.2 Multi Phase Flow Meter

MPFM is a specialized device designed to measure the flow rates of oil, gas, and water as they are produced from a well. MPFM measure the flow rates of different phases simultaneously without separation. MPFM can provide continuous measurement of flow rates at sensor locations with decent accuracy but they are considered extremely expensive and requires regular calibration. Multi-phase flow measurement technology may be an attractive alternative since it enables measurement of unprocessed well streams very close to the well. The use of MPFM may lead to cost savings in the initial installation. However, due to increased measurement uncertainty, a cost-benefit analysis should be performed over the life cycle of the project to justify its application. It operates by determining the individual oil, gas and water phase fractions and their flow rates within a single pipeline or wellbore. MPFM also considered less accurate outside the normal operational range. SPFM are considered more accurate than MPFM since the phases are measured after separation in SPFM ([Meribout et al., 2020](#)).

1.7 Virtual Flow Meters

VFM are virtual model for predicting flow rates. Unlike the physical flow meters, VFM doesn't require deployment of dedicated hardware for flow rate measurement. Both VFM and physical flow meters measure the flow rates indirectly. Physical flow meters bring their own set of sensors whereas VFM use the existing sensors in the well. Over the past two decades, the evolution of VFM has led to the creation of diverse techniques for estimating multi-phase flow rates using field data. This progress has seen the emergence of commercial VFM systems from various companies, widely adopted by oil and gas companies globally. Physical flow meters use physics based mathematical equations to calculate the flow rates whereas VFM can use physics equations and/or ML based models for prediction. VFM, therefore can be categorized into 3 types; physics based, ML based and hybrid flow meters.

1.7.1 Physics-based VFM

Physics based VFM are also known as first principles VFM systems. Physics-based VFMs are grounded in first-principles models that describe the behavior of multiphase fluids flowing through wells and pipelines using the fundamental laws of fluid mechanics, thermodynamics, and heat transfer. These systems predict oil, gas, and water flow rates by solving mass, momentum, and energy conservation equations, typically coupled with industry-standard multiphase flow correlations or mechanistic models to compute pressure gradients, holdup, and flow regimes under varying operating conditions ([Bikmukhametov and Jäschke, 2020b](#); [Ahmed, 2010](#)). Key components of physics-based VFMs include accurate Pressure Volume Temperature (PVT) models to represent phase behavior, inflow and vertical lift performance relationships to describe reservoir-to-surface coupling, and wellbore or pipeline geometries that define boundary conditions. By continuously ingesting real-time measurements such as pressure, temperature, choke position, and wellhead conditions, the VFM iteratively adjusts the multiphase flow solution to match the observed system behavior. This enables the model to reproduce dynamic changes in well performance - such as variations in GOR, water cut, reservoir pressure, or artificial-lift settings without physical multiphase meters. The strength of physics-based VFMs lies in their robustness, as they rely on immutable physical laws rather than purely statistical relationships, making them highly reliable in complex or rapidly changing flow conditions where data-driven models may fail or extrapolate poorly. Following are the main stages of physics-based well modelling used in well performance analysis, production forecasting, and Virtual Flow Metering.

- **Data Acquisition and Model Initialization:** Physics-based well modeling begins with the collection and validation of all essential input data that define

the well's physical and thermodynamic system. This includes completion design (tubing size, deviation, perforation intervals), reservoir properties (pressure, temperature, PI, skin), and PVT data describing the phase behavior of the produced fluids. Operational parameters such as choke size, wellhead pressure, flowing temperature, and artificial-lift settings are also incorporated. The accuracy of the model depends heavily on the quality of this initial data, making data cleaning, consistency checks, and uncertainty quantification critical steps. Once validated, these inputs establish the boundary conditions for subsequent flow calculations.

- **Reservoir Inflow Modeling:** The next stage involves constructing the Inflow Performance Relationship (IPR), which describes how much fluid the reservoir can deliver at different downhole pressures. Depending on reservoir type, engineers may use Vogel, Darcy, Forchheimer, or multiphase inflow models to represent inflow behavior. The IPR curve captures the interaction between reservoir pressure, permeability, fluid PVT characteristics, and near-wellbore conditions such as skin or damage. A robust IPR is essential because it determines the maximum deliverability of the well and forms the upstream boundary condition for all wellbore flow simulations.
- **Vertical and Multiphase Lift Modeling (VLP Curve Generation):** Vertical Lift Performance (VLP) modeling quantifies the pressure required to lift the reservoir fluids from the perforations to the surface. This step uses mechanistic or empirical multiphase flow correlations to compute pressure gradients along the wellbore, considering gas-liquid slip, flow regime transitions, holdup, frictional losses, and hydrostatic head. Factors such as tubing geometry, inclination, heat-transfer conditions, and artificial-lift mechanisms (ESP, gas lift, PCP) strongly influence the VLP results. The resulting VLP curve provides the wellbore pressure response for any given flow rate and forms the critical link between the reservoir and surface system.
- **System Matching (IPR-VLP Intersection):** Once the IPR and VLP curves are generated, the model determines the stable operating point of the well by finding the intersection between the two. This intersection represents the flow rate and bottomhole pressure at which reservoir inflow and wellbore outflow are in equilibrium. System matching ensures that the well model is physically consistent and reflects actual production behavior. Engineers often calibrate this stage using well test data, pressure surveys, or downhole gauges to align the theoretical model with observed performance. Accurate matching is essential for reliable predictions in virtual metering, production optimization, and artificial lift design (Vogel, 1968; Beggs and Brill, 1973).

- **Surface Network and Choke Modeling:** In many wells, surface constraints significantly influence flow performance. This stage incorporates choke models, manifold pressures, flowline hydraulics, separators, and facility back-pressure effects. Choke modeling uses vendor-specific or mechanistic equations to capture the nonlinear relationship between choke opening, upstream pressure, downstream pressure, and multiphase flow rate. Incorporating the surface network allows the model to reflect operational realities, including changing separator pressures, pipeline slugging tendencies, and facility limits, thereby improving steady-state or transient flow predictions.
- **Calibration, Validation, and Sensitivity Analysis:** Physics-based well models must be calibrated against historical production data, well tests, and downhole or surface gauge measurements. Calibration typically involves adjusting uncertain parameters such as PI, skin, GOR, water cut, or tubing roughness until the model matches real observed behavior. Once calibrated, the model is validated by comparing predictions against unseen measurements to ensure predictive reliability. Sensitivity analysis is then performed to quantify how uncertainties in PVT, reservoir pressure, or equipment performance affect model outputs. This step ensures that the model remains robust, stable, and trustworthy for operational use.

Physics based VFM dominate the industry as they've undergone significant development over half a century to comprehensively outline each facet of this approach. This extensive effort has yielded a strong grasp of mechanistic modeling within production systems, fluid properties, and optimization methods. Consequently, physics based VFM stands as a dependable method to depict overall production system behavior and especially multi-phase flow phenomena ([Bikmukhametov and Jäschke, 2020b](#)). Physics based VFM are generally considered expensive due to licensing costs of the simulator and slow in prediction due to processing of high complexity mathematical models. Physics based VFM require very accurate descriptions of the well, fluids, trajectory, production choke and installation parameters for modeling the simulator based on the theories of physics.

Physics-based VFMs can be directly compared across several important technical dimensions that determine their suitability for different production environments. One of the most important aspects is the complexity of the model, which varies significantly between vendors. Some VFMs, such as SLB's OLGA VM and Aker's LiveVFM, use transient multiphase flow simulations, which allow them to capture dynamic behaviors such as slugging, gas breakthrough, and changing Gas-Oil Ratio (GOR). These tools are preferred for deepwater subsea wells, long tiebacks, and high-rate gas condensate production systems. Others, such as Petex PROSPER/GAP, Tieto Energy Components VFM, and BGS VMS, use steady-state mechanistic models, which are far simpler,

computationally efficient, and suitable for most onshore wells or stable offshore production.

1.7.2 Machine Learning VFM or Data Driven VFM

Machine Learning (ML) based modeling involves analyzing system data to uncover connections between input and output variables without precise knowledge of the system's physical behavior. ML based VFM are purely data driven. This method proves advantageous in bypassing intricate physical modeling, especially for systems like multi-phase flows in pipes, where exact numerical solutions can be challenging. It relies on experimental or industrial data to grasp the system's behavior directly and attempts to learn the underlying relationships from this data. ML based VFM modeling involves collecting data such as downhole and wellhead pressures and temperatures, choke regulation levels and corresponding oil, gas, and water flow-rate measurements (Agwu et al., 2022). These measurements can originate from various sources in the form of well-test data or MPFM. The ML based model can serve as a backup metering system for individual wells if MPFM are installed for each wellhead. The ML based model functions as a standalone VFM system.

1.7.3 Hybrid VFM

Hybrid VFM combines the physics based and ML based models to improve the performance of the system. All 3 types of VFM rely on pressure and temperature sensor data in the well. ML based VFM and hybrid VFM also rely on physical well-test measurements for model training. More the well-test measurements available, higher the accuracy of the model within the operating conditions of the well-test measurements. They are however said to have higher degree of uncertainty outside the operating conditions of the well-test measurements. All types of VFM have the limitation of being restricted to steady state, meaning that if there are rapid transient changes in the flow, VFM will not predict them accurately. Therefore VFM cannot predict startup, shut-down and other transient scenarios. ML and hybrid VFM also cannot accurately handle advanced what-if predictive scenarios. Physics based VFM in contrast can accurately handle advanced what-if predictive scenarios.

1.8 Research Problem

Accurate estimation of multiphase flow rates in producing oil and gas wells is a fundamental requirement for effective production monitoring, reservoir management, and operational decision-making. Traditionally, this task is performed using MPFMs, which provide direct measurements of oil, gas, and water flow rates. However, MPFMs are capital-intensive, require regular calibration and maintenance, and are often deployed

at limited locations due to cost and operational constraints. As a result, continuous and comprehensive flow rate measurements are rarely available across all wells and operating conditions.

To address these limitations, VFM techniques have been developed as software-based alternatives to physical flow meters. The conventional VFM approaches rely on first-principles physics-based models, which use pressure, temperature, choke settings, and fluid properties to estimate flow rates. While these models are grounded in physical laws and offer interpretability, their accuracy is often limited by simplifying assumptions, uncertainty in model parameters, unmodeled multiphase flow phenomena, and changing well and reservoir conditions over time. Consequently, purely physics-based VFMs may exhibit systematic bias or degraded performance when applied outside their calibration envelope.

Recent advances in data-driven and machine learning techniques have demonstrated strong predictive capabilities for complex nonlinear systems, including multiphase flow behavior. However, purely data-driven models typically require large volumes of high-quality labeled data, which are seldom available in production environments due to sparse MPFM measurements and irregular well testing schedules. Furthermore, black-box ML models often lack physical interpretability and may generate non-physical predictions when extrapolated beyond the training data, limiting their acceptance in safety-critical industrial applications.

This situation highlights a fundamental challenge in Virtual Flow Metering: physics-based models provide interpretability ([Karniadakis et al., 2021](#)) and physical consistency but suffer from model inaccuracies, while data-driven models offer flexibility and expressive power but are constrained by limited data availability and a lack of physical grounding. Neither approach alone provides a fully satisfactory solution under realistic industrial conditions characterized by sparse supervision, noisy measurements, and evolving operating regimes.

A promising approach to overcome these limitations is the development of hybrid or physics-informed learning architectures, in which physical models and machine learning are combined in a complementary manner. In particular, *residual learning* offers a structured framework whereby a physics-based model produces a baseline estimate of flow rates, and a data-driven component learns to model the residual errors arising from unmodeled dynamics, parameter uncertainty, and measurement noise. This formulation preserves physical interpretability while enabling data-driven correction of systematic model deficiencies.

Despite increasing interest in hybrid modeling approaches, there remains a lack of systematic investigation into physics-informed residual learning architectures for Virtual Flow Metering under sparse and irregular measurement conditions, particularly in multi-well production systems. Key open questions include how residual models can be trained effectively with limited ground-truth flow measurements, how physics-

generated states can be leveraged as informative context without introducing data leakage, and how such architectures compare against traditional physics-only and purely data-driven methods in terms of accuracy, robustness, and generalization. Therefore, the research problem addressed in this thesis can be stated as follows:

How can a physics-informed residual learning architecture be designed and evaluated to improve the accuracy and robustness of virtual flow metering in oil and gas wells under sparse and irregular flow rate measurements?

This research aims to bridge the gap between physics-based modeling and data-driven learning by proposing and validating a hybrid framework that exploits the strengths of both paradigms while adhering to the practical constraints of industrial oil and gas production systems.

1.9 Research Objectives

The primary objective of this research is to develop and evaluate a physics-informed residual learning framework for virtual flow metering in oil and gas production wells under conditions of sparse and irregular flow rate measurements. To achieve this overarching goal, the following specific research objectives are defined:

1. **To review and analyze existing virtual flow metering approaches**, including conventional physics-based models, purely data-driven methods, and hybrid modeling techniques, with particular emphasis on their strengths and limitations in industrial production environments.
2. **To formulate a baseline physics-based virtual flow metering model** that estimates multiphase flow rates using routinely available well measurements such as pressures, temperatures, and choke settings, and to assess its performance and limitations under varying operating conditions.
3. **To design a physics-informed residual learning architecture** in which a data-driven model is trained to learn and correct the residual errors of the baseline physics-based flow model, thereby improving prediction accuracy while preserving physical interpretability.
4. **To develop a data preprocessing and feature engineering strategy** that enables effective training of the residual learning model under sparse and irregular ground-truth flow measurements, including the integration of physics-generated state variables as informative contextual inputs.

5. **To develop a computationally efficient prediction framework** that enables scalable virtual flow metering across large well populations, recognizing that industrial oil and gas assets may consist of thousands of wells distributed across multiple fields and production systems.
6. **To implement the proposed hybrid modeling architecture** using real-world multi-well production data and to ensure that the training, validation, and testing procedures avoid data leakage and reflect realistic operational constraints.
7. **To evaluate the performance of the proposed physics-informed residual learning approach** using appropriate regression metrics, and to compare its predictive accuracy against physics-only and purely data-driven baseline models.
8. **To analyze the robustness and generalization capability of the proposed framework** across multiple wells and operating regimes, with particular attention to its behavior under limited training data and changing system conditions.
9. **To assess the practical applicability of the proposed approach** for industrial virtual flow metering by examining its interpretability, computational efficiency, and potential integration into existing production monitoring workflows.

1.10 Scope and Limitations

This research focuses on the development and evaluation of a physics-informed residual learning architecture for virtual flow metering in oil and gas production wells. The scope of the study and its inherent limitations are outlined below to clearly define the boundaries of the work and to contextualize the results.

1.10.1 Scope of the Research

1. The study is confined to *producing oil and gas wells* equipped with standard surface and downhole instrumentation, including pressure, temperature, and choke measurements, which are routinely available in production operations.
2. The proposed virtual flow metering framework integrates a *first-principles physics-based model* with a *data-driven residual learning component* to estimate multiphase flow rates, with particular emphasis on oil, gas, and water production rates.
3. The research considers *sparse and irregular ground-truth flow measurements*, such as those obtained from periodic well tests, reflecting realistic industrial data availability constraints.

4. The evaluation of the proposed approach is conducted using *historical production data from multiple wells*, enabling assessment of the model's performance, robustness, and generalization capability across different wells and operating conditions.
5. Model performance is assessed using standard regression metrics, including accuracy and error-based measures, and is compared against baseline physics-only and data-driven modeling approaches.
6. The study emphasizes *software-based virtual flow metering* and does not consider hardware design, sensor deployment strategies, or real-time control system implementation.

1.10.2 Limitations

1. The accuracy of the proposed framework is inherently dependent on the quality, availability, and representativeness of the input data, including sensor measurements and ground-truth flow rate observations.
2. The baseline physics-based model employed in this study involves simplifying assumptions regarding multiphase flow behavior, fluid properties, and well conditions, which may not fully capture complex transient or highly nonlinear phenomena.
3. The data-driven residual learning component is trained on historical data and may exhibit reduced performance when applied to operating regimes or well conditions that are significantly different from those observed during training.
4. The proposed framework is evaluated using offline historical datasets and does not explicitly address real-time deployment challenges such as computational latency, online model updating, or integration with supervisory control and data acquisition (SCADA) systems.
5. The study does not explicitly quantify uncertainty in flow rate predictions, such as confidence intervals or probabilistic uncertainty bounds, although this represents a potential direction for future research.
6. The findings of this research are specific to the datasets and wells considered and may require recalibration or retraining before application to different reservoirs, fields, or asset types.

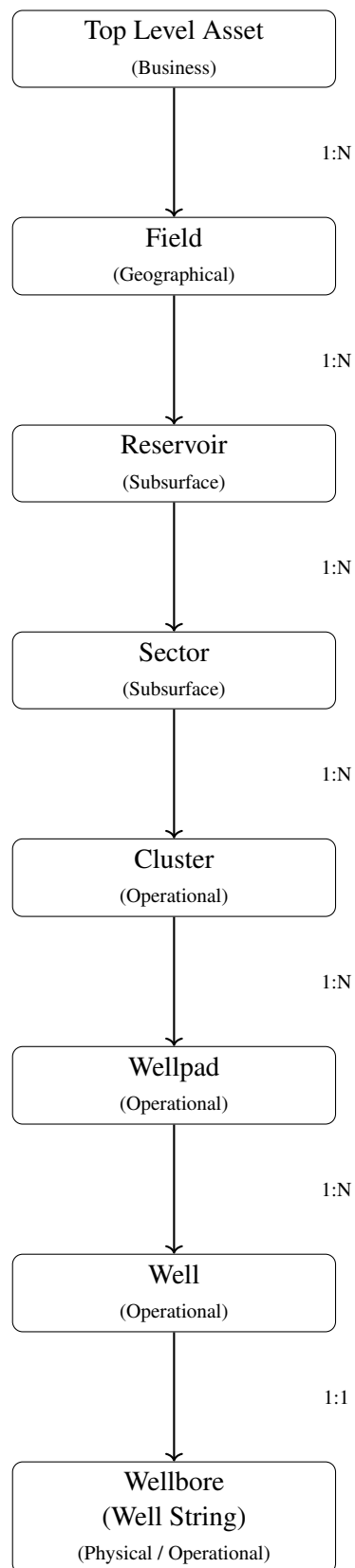


Figure 1.1
Oil and Gas Field Asset Hierarchy

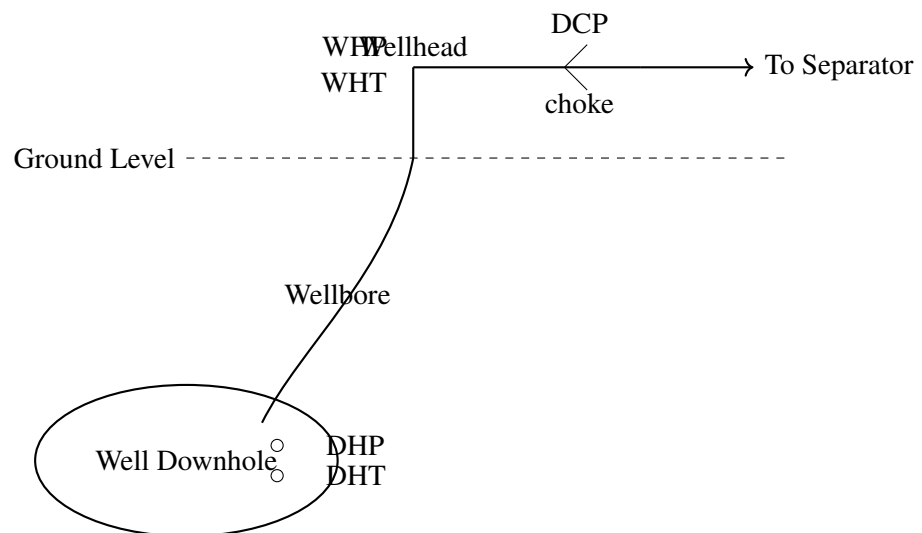


Figure 1.2
Well Downhole and Surface Instrumentation

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The accurate estimation of oil, gas, and water production rates is a fundamental requirement in petroleum production engineering, underpinning reservoir management, production optimization, flow assurance, and economic decision-making. Traditionally, phase flow rates are measured using physical multiphase flow meters (MPFMs). While MPFMs provide direct measurements, their widespread deployment is limited by high capital cost, maintenance requirements, calibration complexity, and installation constraints, particularly in offshore, subsea, and marginal field developments ([Corneliussen et al., 2005](#); [Meribout et al., 2020](#); [Hansen et al., 2019](#)).

As a cost-effective alternative, Virtual Flow Metering (VFM) has emerged as a key enabling technology. VFM systems estimate phase flow rates indirectly using routinely available surface and downhole measurements such as pressures, temperatures, and choke settings. Over the past two decades, VFM methodologies have evolved from purely physics-based models to data-driven machine learning approaches and, more recently, to hybrid physics-data architectures ([Bikmukhametov and Jäschke, 2020b](#)). This chapter reviews these developments, critically examines their strengths and limitations, and establishes the motivation for a physics-informed hybrid residual learning architecture for virtual flow metering.

2.2 Physics-Based Virtual Flow Metering

2.2.1 Foundations of Petroleum Production Modeling

Physics-based VFM approaches rely on first-principles models derived from reservoir engineering, fluid mechanics, and thermodynamics. Classical petroleum engineering formulations describe the relationship between reservoir pressure, flowing bottom-hole pressure, and production rate using inflow performance relationships (IPRs) ([Dake and Dake, 1978](#)). These formulations are widely used in nodal analysis to predict well performance under steady-state conditions.

IPR models typically assume homogeneous reservoir properties, steady-state flow, and known fluid characteristics. While these assumptions are reasonable during early production or controlled testing, they are frequently violated during long-term field operation, leading to systematic prediction errors.

2.2.2 Choke Flow and Surface Constraints

Surface choke flow models form a critical component of physics-based VFMs by linking upstream pressure, downstream pressure, and choke opening to flow rates. Numerous empirical and semi-empirical correlations have been developed to model multiphase flow through chokes (Agwu et al., 2022; Alrumah and Alenezi, 2019). These correlations are often calibrated for specific flow regimes and fluid compositions.

Although choke models provide physically interpretable relationships, their accuracy is highly sensitive to operating conditions. Flow regime transitions, changing gas-liquid ratios, and water breakthrough can significantly degrade predictive performance. Consequently, choke correlations often require frequent recalibration, limiting their robustness for long-term monitoring.

2.2.3 Strengths and Limitations of Physics-Based VFMs

Physics-based VFMs offer several advantages, including interpretability, compliance with physical laws, and the ability to extrapolate beyond observed data ranges. However, they suffer from key limitations. Uncertain reservoir parameters, incomplete representation of multiphase flow physics, and measurement noise can lead to biased predictions (Bikmukhametov and Jäschke, 2020b). Moreover, mechanistic models struggle to capture complex nonlinear interactions between operational variables, motivating the exploration of data-driven alternatives.

2.3 Data-Driven Virtual Flow Metering

2.3.1 Early Machine Learning Approaches

The application of machine learning to VFM began with artificial neural networks (ANNs) and regression-based soft sensors (Al-Qutami et al., 2017; AL-Qutami et al., 2017). These models learned direct mappings between sensor measurements and phase flow rates, often outperforming physics-based models under stable operating conditions.

However, early ANN-based VFMs exhibited limited generalization capability and were prone to overfitting, particularly when training datasets were small or unevenly distributed across operating regimes.

2.3.2 Advanced Data-Driven Models

Subsequent research explored ensemble learning, adaptive optimization techniques, and Bayesian neural networks to improve robustness and uncertainty quantification (AL-Qutami et al., 2018; Grimstad et al., 2021). Ensemble methods reduced variance

by combining multiple learners, while Bayesian approaches provided probabilistic estimates of prediction uncertainty. (Grimstad et al., 2021) develops a ML based VFM, implemented on Bayesian Neural Network (BNN). BNN provide a probabilistic distribution over weights and predictions unlike traditional neural networks, which produce point estimates for weights and predictions. They have trained the model on a large and heterogeneous data-set, consisting of 60 wells across five different oil and gas assets. The predictive performance is analyzed on historical and future test data, where an average error of 4%–6% and 8%–13% is achieved for the 50% best performing models, respectively.

More recently, deep learning architectures have been applied to VFM, demonstrating strong predictive performance in data-rich environments (Song et al., 2022; Mercante and Netto, 2022). Despite these advances, purely data-driven VFMs remain fundamentally dependent on large, high-quality datasets and often lack physical interpretability.

(Muchsin et al., 2023) proposed a VFM model using a time series-time delay artificial neural network technique in predicting the multi-phase flow rate accurately with the best average discrepancy of Q_{gas} (6.0%), Q_{oil} (-16.4%), and Q_{water} (-2.4%) compared with actual measurement by MPFM. The training of the VFM model, which takes less than 7 minutes and has acceptable values for MSE, R, and MAPE, demonstrates that this method is reliable enough to be used in the oil and gas industries, which often require conducting dozens of individual well tests in their daily activities. The utilization of data measurement of existing well orifice meter combined with data measurement from choke valve and well head as parameter data input of the network has proven effective to improve the accuracy of the VFM model. In addition, the greater the number of data used in training phase determines the accuracy performance of the VFM model. However, concerning that this study only uses limited well reference data, it is recommended to implement this VFM model on other wells in different fields to validate its performance accuracy during well testing.

(AL-Qutami et al., 2018) introduces a VFM system employing ensemble learning tailored for fields with limited data from a shared metering setup. The method creates diverse neural network learners by manipulating training data, neural network structure, and learning approach. Using Adaptive Simulated Annealing (ASN) optimization, it selects the best subset of learners and an optimal combining strategy. Assessment using real well test data shows exceptional performance, with average errors of 4.7% and 2.4% for liquid and gas flow rates respectively. The accuracy of their VFM was further confirmed through cumulative deviation analysis, where predictions fall within a maximum deviation of $\pm 15\%$. Comparisons with standard bagging and stacking techniques demonstrate significant enhancements in both accuracy and ensemble size. The proposed VFM system offers ease in development and maintenance compared to traditional model-driven VFM, requiring only well test samples for model

tuning. It is anticipated that this developed VFM can complement physical meters, improve data consistency, aid in reservoir management and flow assurance, ultimately resulting in more efficient oil recovery and reduced operational and maintenance costs.

(AL-Qutami et al., 2017) introduces a radial basis function network designed to create a virtual flow meter (VFM) specifically for estimating gas flow rates within multi-phase production lines. Validating the model with real well test data demonstrates its outstanding performance and ability to generalize effectively. Furthermore, the paper delves into the importance of downhole and choke valve measurements in ensuring precise predictions. This proposed VFM model presents a potentially appealing and cost-efficient solution for real-time production monitoring needs while simultaneously reducing operational and maintenance expenses.

In a separate work also by (Al-Qutami et al., 2017) proposes a method for estimating phase flow rates in oil and gas production wells using readily available measurements. By overcoming the limitations of traditional metering facilities, their system offers a cost-effective way to monitor production in real time. It not only cuts operational and maintenance expenses but also serves as a reliable backup to multi-phase flow meters. The technique involves creating a soft-sensor through a feed-forward neural network. To ensure accuracy without excessive complexity, the system uses K-fold cross-validation and an early stopping technique to regulate generalization and network intricacy. Validation using real well test data demonstrates the sensor's effectiveness, evaluated through cumulative deviation and flow plots. Their results indicate promising performance, with an average error of approximately 4% and less than 10% deviation for 90% of the samples.

(Song et al., 2022) employs Back Propagation Neural Network (BPNN), LSTM network, and Random Forest (RF) algorithm to develop an intelligent ML based model for virtual flow meters in oil and gas development. Their data-set is constructed using actual data from two oil wells in an offshore oil field in the South China Sea. Among the three models, the LSTM model has demonstrated the highest accuracy, with a Mean Absolute Error (MAE) of 3.9%. LSTM has also demonstrated the highest stability and requires a moderate amount of data volume. BP network on the other hand have exhibited the lowest accuracy, with a MAE of 12.1%, as well as the lowest stability. BP however has shown the smallest data volume requirement. The Random Forest model has shown moderate accuracy, high stability, but has required the highest data volume.

2.3.3 Time-Series and Attention-Based Models

The increasing availability of time-stamped production data has motivated the use of time-series models in VFM. Recurrent neural networks and long short-term memory (LSTM) networks have been used to capture temporal dependencies in production be-

havior ([Andrianov, 2018](#)). This is because unlike feed-forward neural network, which process input data in a one-way, LSTM-RNN are designed to process sequential data, such as time series. ([Andrianov, 2018](#)) has achieved the best accuracy when the lengths of the input and output sequences to LSTM are equal.

2.3.4 Limitations of Purely Data-Driven VFMs

Despite strong predictive performance, purely data-driven VFMs exhibit fundamental limitations. They lack explicit physical grounding, struggle with extrapolation beyond the training domain, and may violate physical constraints such as non-negativity or phase consistency ([Bikmukhametov and Jäschke, 2020b](#)). The black-box nature of deep learning models also reduces trust and hinders adoption in industrial decision-making contexts.

2.4 Hybrid Physics - Data Virtual Flow Metering

2.4.1 Motivation for Hybrid Modeling

Hybrid modeling approaches seek to combine the strengths of physics-based and data-driven models. In VFM applications, physics-based models capture dominant flow behavior, while machine learning models excel at learning unmodeled nonlinearities and systematic discrepancies ([Hotvedt et al., 2020](#)). The hybrid VFM proposed by ([Ishak et al., 2022](#)) uses ensemble learning to develop the data driven model by incorporating multiple ML models. The data driven model is then combined with the physics model using a combiner. They have achieved a 50% improvement in performance using the combiner compared to their stand-alone performance of physics based and ML based VFM.

([Nemoto et al., 2023](#)) introduces a cloud-based VFM that combines physics and data techniques to accurately estimate water production per well within a gas field. By integrating physics-based models tailored for high gas volume fraction gas-liquid flows in the wellbore and employing a data-driven approach to fine-tune these models with actual well test data, this hybrid method ensures precise real-time estimations. Its adaptability to evolving well performance and increased water production creates a self-calibrating solution, ensuring ongoing accuracy and relevance despite changing production and well conditions. Their VFM system demonstrates substantial alignment with well test data under steady-state conditions, affirming its reliability. Operating remotely through a cloud-based Cognite DataOps platform, this system conducts calculations and stores results, facilitating continual access to live sensor data for other applications or visualization via a web interface. Utilizing existing sensors within the wells, the VFM system offers cost efficiency by minimizing both initial investment and operational expenses in comparison to installing multi-phase flow meters or separators.

Hybrid VFM models have become increasingly popular since 2018 and increase usage of neural networks seen for developing both physics and ML based models.

2.4.2 Residual Learning Frameworks

Residual learning has emerged as a particularly effective hybrid strategy. In this framework, a physics-based model provides baseline predictions, and a machine learning model is trained to learn residual errors between predictions and measurements (Bikmukhametov and Jäschke, 2020a). This separation improves data efficiency, stabilizes training, and preserves physical interpretability.

Residual hybrid VFMs have demonstrated improved accuracy and robustness compared to both purely physics-based and purely data-driven approaches (Ishak et al., 2022; Gryzlov et al., 2023).

2.4.3 Industrial Applications

Hybrid VFMs have been successfully deployed in industrial environments, including cloud-based real-time monitoring systems for offshore fields (Nemoto et al., 2023). These studies highlight the scalability and operational relevance of hybrid architectures, particularly in settings with sparse instrumentation and limited well-test data.

2.5 Physics-Informed Machine Learning

2.5.1 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) incorporate governing equations directly into the training process and have been explored for flow metering applications (Franklin et al., 2022). While PINNs offer a principled approach to enforcing physical laws, their application to multiphase well flow remains challenging due to incomplete or uncertain governing equations. (Franklin et al., 2022) proposes a hybrid VFM combining Physics Informed Neural Networks (PINN) based physics model and LSTM-RNN based ML model. Their resulting hybrid model is capable of predicting the average flow rate some time-steps ahead using the measurements available in the oil well information system. Unlike the ML based LSTM-RNN model proposed by (Andrianov, 2018), the system proposed by (Franklin et al., 2022) is hybrid model using neural networks for physics based model as well as the ML based model.

2.5.2 Soft Physics Constraints and Closure Laws

An alternative to strict physics enforcement is the use of soft physical constraints embedded in model architecture. Examples include bounded logistic water-cut closures and explicit non-negativity constraints on phase rates (Bikmukhametov and Jäschke,

2020b). These approaches improve physical realism while retaining flexibility and numerical stability.

2.6 Rationale for a Physics-Informed Hybrid Residual Architecture

The adoption of a physics-informed hybrid residual architecture in this work is motivated by several well-established considerations in the virtual flow metering and data-driven modeling literature:

- **Limited availability and sparsity of well test data:** In practical production environments, well test measurements are acquired infrequently due to operational constraints, cost, and production priorities. Purely data-driven models typically require large, dense, and regularly sampled datasets to generalize reliably, which is rarely achievable in field conditions. Physics-based models, by contrast, can operate under sparse data regimes by embedding prior physical knowledge, making them a natural foundation for hybrid approaches.
- **Strong prior knowledge of multiphase flow behavior:** The governing relationships between pressure, temperature, fluid properties, and phase flow rates in production systems are well understood and extensively documented in the petroleum engineering literature. Ignoring this prior knowledge and learning flow behavior purely from data is inefficient and may lead to physically implausible predictions. A physics-informed architecture ensures that baseline predictions remain consistent with known flow physics.
- **Systematic bias and model mismatch in physics-based models:** While physics-based virtual flow metering models capture dominant flow mechanisms, they often rely on simplifying assumptions, empirical correlations, and uncertain parameters. As a result, systematic residual errors arise due to unmodeled effects such as changing flow regimes, sensor drift, fluid property variations, and facility interactions. Learning residual corrections allows these structured biases to be addressed without discarding the physical model.
- **Improved generalization across wells and operating conditions:** Residual learning focuses the machine learning component on modeling deviations from physics rather than the full input-output mapping. This reduces model complexity and improves generalization, particularly when transferring predictive capability across wells with heterogeneous characteristics. Such decomposition has been shown to enhance robustness under spatial and temporal distribution shifts.

- **Reduced risk of overfitting and improved data efficiency:** By constraining the learning task to residual errors, the hybrid architecture significantly reduces the hypothesis space explored by the machine learning model. This improves data efficiency and mitigates overfitting, which is critical when training on small multi-well datasets typical of well test applications.
- **Interpretability and diagnostic value:** Separating physics-based predictions from learned residual corrections enhances model interpretability. The physics component provides transparent baseline behavior, while the residual component highlights conditions under which the physical assumptions break down. This separation supports diagnostic analysis and builds trust for deployment in production engineering workflows.
- **Alignment with operational calibration practices:** In field operations, physics-based models are routinely calibrated using limited well-specific data, while global correlations and assumptions are reused across assets. The proposed hybrid architecture mirrors this practice by allowing limited per-well calibration of physical parameters while maintaining a globally trained residual model, thereby aligning model design with real-world engineering workflows.
- **Robustness under non-stationary operating conditions:** Production systems evolve over time due to changes in reservoir conditions, choke settings, artificial lift performance, and facility constraints. Hybrid residual models can adapt to such non-stationarities by updating residual behavior while retaining physically consistent baseline predictions, offering greater long-term stability than purely data-driven approaches.

Table 2.1

Comparative evaluation of flow metering and virtual flow metering approaches

Criterion	MPFM	Physics-Based VFM	ML-Based VFM	Hybrid Physics-ML VFM
Measurement principle	Direct physical measurement using nuclear, differential pressure, impedance, or venturi-based sensors	Governing mass, momentum, and energy balance equations with empirical closure relations	Statistical or machine-learning-based mapping between sensor inputs and flow rates	Physics-guided data-driven inference combining physical laws and machine learning
Accuracy under ideal conditions	High when calibrated and operated within design envelope	Moderate to high, depending on model fidelity and calibration	High within training data domain	High across wide operating ranges due to physics-guided learning
Performance under changing flow regimes	Degrades under slugging, emulsions, and severe regime transitions	Sensitive to regime assumptions and closure law validity	Poor extrapolation beyond trained regimes	Robust due to physical consistency and adaptive correction
Data re-requirements	Minimal (sensor signals only)	Low to moderate (well geometry, PVT properties, boundary conditions)	High (large, labeled datasets required)	Moderate (limited labeled data supplemented by physics constraints)

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Table 2.1 – continued

Criterion	MPFM	Physics-Based VFM	ML-Based VFM	Hybrid Physics-ML VFM
Dependence on well tests	Required periodically for recalibration	Strong dependence for parameter estimation	Strong dependence for supervised learning	Reduced dependence due to physics-informed training
Generalization capability	Limited to hardware operating envelope	Limited across wells due to well-specific parameters	Limited; prone to overfitting	Improved cross-well and cross-condition generalization
Physical interpretability	High (direct physical measurements)	High (explicit physical meaning of parameters)	Low (black-box behavior)	High to moderate with interpretable residual structure
Computational cost	Low during operation; high installation and maintenance cost	Moderate due to nonlinear solvers	Low inference cost; high training cost	Moderate; suitable for near real-time deployment
Installation and operational cost	Very high (CAPEX and OPEX)	Low (software-based)	Low (software-based)	Low (software-based)
Sensitivity to sensor noise	Moderate; hardware filtering required	High sensitivity due to noise propagation through equations	High unless explicitly regularized	Reduced through physics-based regularization

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Table 2.1 – continued

Criterion	MPFM	Physics-Based VFM	ML-Based VFM	Hybrid Physics-ML VFM
Scalability across fields	Poor due to hardware deployment constraints	Moderate with per-well calibration	High if sufficient data exists	High with minimal re-calibration
Long-term maintainability	Challenging due to sensor drift and failures	Moderate; requires periodic recalibration	Challenging due to model drift and retraining	Improved stability via physics constraints

Table 2.2
Literature summary

Research	VFM Type	ML Model	Evaluation Metrics
AL-Qutami et al. (2017)	ML	ANN	Oil) $R^2 = 0.965$, $RMSE = 1.24899$, $MAPE = 4.22$ Gas) $R^2 = 0.954$, $RMSE = 1.35277$, $MAPE = 2.27$
Al-Qutami et al. (2017)	ML	RBF	$R^2 = 0.93978$, $RMSE = 1.334$, $MAPE = 6.16$
Andrianov (2018)	ML	LSTM, RNN	Not reported
AL-Qutami et al. (2018)	ML	Ensemble and ASN	NN $ANN : RMSE = 0.0585$; $STDEV = 0.0046$; $MAPE = 4.7$ $ASA : RMSE = 0.0442$, $STDEV = 0.0036$, $MAPE = 2.35$
Dutta and Kumar (2018)	ML	ANN FPA	$RMSPPE = 0.75$, $ARPE = 99.25$
Hansen et al. (2019)	ML	ANN	a) $R = 0.993$, $AAPE = 8.39$ b) $R = 0.995$, $AAPE = 6.36$

Table 2.2 – continued

Research	VFM Type	ML Model	Evaluation Metrics
Alrumah and Alenezi (2019)	ML	ANN, LSSVM, SIMPLEX	$ANN : R^2 = 0.9292, RMSE = 863.98, AARPE = 22.06$ $LSSVM : R^2 = 0.9477, RMSE = 719.6, AARPE = 21.5$ $SIMPLEX : R^2 = 0.885, RMSE = 1067, AARPE = 26.8$
Bikmukhametov and Jäschke (2020a)	ML	MLP, LSTM, GB	$Oil : MLP : RMSE = 0.0458; LSTM : RMSE = 0.0476$ $GB : RMSE = 0.0463$ $Gas : MLP : RMSE = 0.0328, LSTM : RMSE = 0.278$ $GB : RMSE = 0.0367$
Hotvedt et al. (2020)	Hybrid	Not reported	$RMSE = 15, MAE = 8$
Marfo and Kporxah (2021)	ML	ANN	$R = 0.9966, MAPE = 3.18$
Grimstad et al. (2021)	ML	BNN	
Ishak et al. (2022)	Hybrid	Ensemble model	
Song et al. (2022)	ML	BPNN, LSTM, Random Forest	
Franklin et al. (2022)	Hybrid	PINN, LSTM, RNN	$10^{-4} \leq MSE \leq 10^{-3}$
Muchsin et al. (2023)	ML	TSTD-ANN	$MSE \leq 1500, R^2 \geq 0.95, MAPE \leq 65$

Table 2.2 – continued

Research	VFM Type	ML Model	Evaluation Metrics
Nemoto et al. (2023)	Hybrid	Linear Regression	M1) $MRPE = 12.64, MAPE = 20.61, R^2 = 0.77$ M2) $MRPE = 0.43, MAPE = 17.43, R^2 = 0.80$ M3) $MRPE = -0.23, MAPE = 20.10, R^2 = 0.64$

CHAPTER 3

METHODOLOGY

3.1 Overview of the Proposed Architecture

This thesis proposes a physics-informed residual learning architecture for virtual flow metering that combines first-principles physical modeling with data-driven machine learning to estimate multiphase flow rates in oil and gas production wells. The framework is designed to exploit the complementary strengths of physics-based models and data-driven approaches while addressing the practical constraints of sparse and irregular flow rate measurements typically encountered in industrial environments.

The overall architecture consists of three main components:

1. **A baseline physics-based flow model**, calibrated independently for each well, which provides first-order estimates of oil, gas, and water flow rates based on physically interpretable relationships and routinely available well measurements.
2. **A global data-driven residual learning model**, trained across multiple wells to learn the systematic residual errors between physics-based predictions and measured flow rates, thereby capturing unmodeled dynamics and nonlinear effects.
3. **A hybrid inference mechanism**, which combines the physics-based predictions with the learned residual corrections under physically motivated gating rules to ensure robustness, physical plausibility, and stable performance under varying operating conditions.

A schematic representation of the architecture is conceptually illustrated in Figure 3.1.

3.1.1 Baseline Physics-Based Inflow Modeling

At the core of the proposed framework lies a physics-based virtual flow metering model that provides first-order estimates of oil, water, and gas flow rates. This model is formulated using simplified physical relationships that capture the dominant dependencies between flow rates and routinely available well measurements, including downhole pressure, choke opening, downstream pressure, and temperature measurements. The physics model incorporates:

- A pressure-driven liquid inflow relationship governed by an effective reservoir pressure

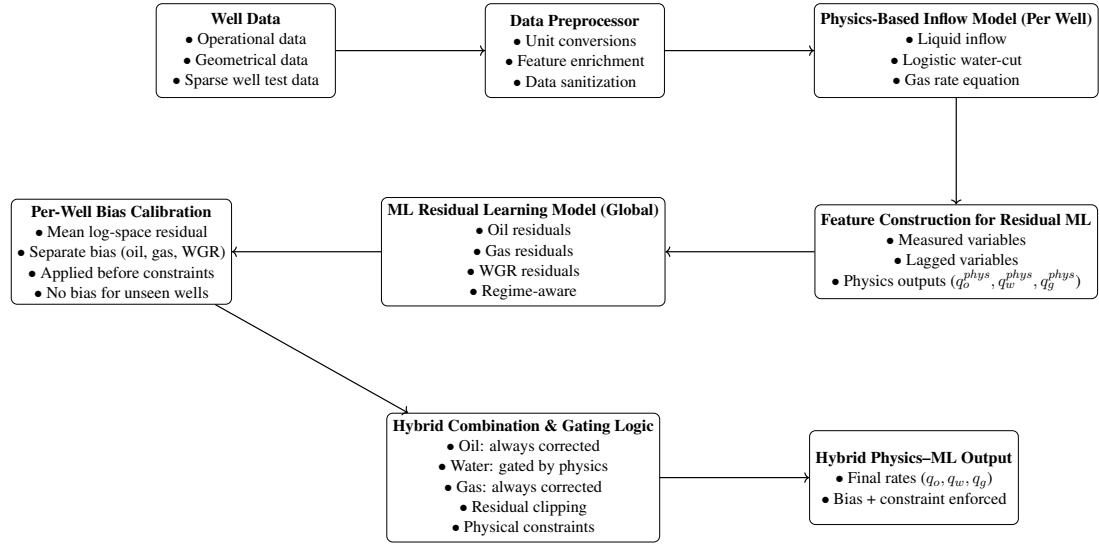


Figure 3.1
Hybrid Physics-Informed Residual Learning Architecture for Virtual Flow Metering

- A water-cut formulation expressed through a logistic function of pressure, temperature, and choke-related features
- A gas flow component modeled as a pressure-drop-driven relationship modulated by choke effectiveness

Model parameters are calibrated on a per-well basis using nonlinear least-squares optimization against available multiphase flow meter measurements. This calibration step ensures that the physics model remains interpretable, physically consistent, and tailored to individual well characteristics while avoiding excessive model complexity.

3.1.2 Residual Learning using Machine Learning

While the physics-based model captures the dominant flow mechanisms, it is inherently limited by simplifying assumptions, parameter uncertainty, and unmodeled multiphase flow effects. To address these deficiencies, a data-driven residual learning component is introduced.

Rather than predicting flow rates directly, the machine learning model is trained to learn the residual error between measured flow rates and physics-based predictions. Residuals are formulated in logarithmic space to stabilize variance and to ensure physically meaningful corrections, particularly at low flow rates. The residual model:

- Operates globally across all wells to leverage shared flow behavior
- Uses both measured operational variables and physics model predictions as input features

- Implemented using a gradient-boosted ensemble regressor capable of modeling nonlinear interactions while maintaining robustness under limited data

Lagged pressure features are incorporated to provide limited temporal context without imposing strong sequential assumptions, enabling the model to capture delayed system responses while remaining compatible with sparse measurement regimes.

3.1.3 Hybrid Inference and Physics-Guided Gating

During inference, physics-based predictions and machine-learned residuals are combined to produce final flow rate estimates. Importantly, residual corrections are not applied indiscriminately. Instead, physically motivated gating mechanisms are employed to ensure that machine learning corrections are only activated under conditions where the physics model is expected to be unreliable. For example:

- Oil rate corrections are selectively applied under high water-cut or large residual conditions
- Water rate corrections are activated only when water production exceeds a minimum threshold
- Gas rate predictions are consistently corrected due to higher uncertainty in gas modeling under varying choke and pressure conditions

These gating strategies preserve physical plausibility, prevent non-physical predictions, and enhance model robustness when extrapolating beyond the calibration range.

3.1.4 Multi-Well Structure and Model Deployment

The proposed framework is designed to support scalable deployment across multiple wells by combining well-specific physics modeling with a globally trained residual learning component. For each well, the physics-based model is calibrated independently to capture well-specific characteristics such as reservoir pressure, inflow behavior, and operational conditions. In contrast, the machine learning residual model is trained across data from all wells, enabling it to learn shared patterns in the discrepancies between physics-based predictions and observed well-test measurements.

This hybrid structure allows the model to retain physically interpretable behavior at the well level while leveraging cross-well information to improve predictive accuracy. As demonstrated in both temporal and spatial generalization experiments, the residual model captures transferable correction patterns that generalize effectively to unseen wells under the Leave-One-Well-Out (LOWO) evaluation framework.

For deployment, the framework supports prediction based on available input measurements such as pressure, temperature, and choke settings. To facilitate practical

implementation, the framework includes model persistence capabilities that allow calibrated physics parameters and trained residual models to be stored and reused. This enables efficient deployment in operational environments, where models can be initialized using historical data and applied to new wells with minimal additional calibration.

3.1.5 Key Advantages of the Proposed Architecture

The proposed physics-informed residual learning architecture offers several key advantages:

- Improved accuracy over physics-only virtual flow metering by compensating for systematic model errors
- Improved robustness and physical consistency compared to purely data-driven models
- Suitability for sparse and irregular measurement regimes
- Interpretability through explicit separation of physics-based predictions and data-driven corrections
- Scalability to multi-well production systems

Together, these characteristics make the proposed framework a practical and scientifically grounded solution for virtual flow metering in real-world oil and gas production environments.

3.2 Dataset Description and Preprocessing

The dataset used in this research consists of multi-source production and operational measurements collected from a set of producing gas wells. The data integrate well test measurements, MPFM data, and permanently installed surface and downhole sensor readings. This study uses data from the same offshore gas field as (Nemoto et al., 2023), although the wells are not identical, still allowing for a meaningful comparison under similar field conditions. A defining characteristic of the dataset is that **well test data are sparse and irregular in time**. Well tests are conducted at non-uniform intervals determined by operational, logistical, and economic considerations rather than by systematic sampling schemes. Consequently, the temporal spacing between successive well tests may range from days to several weeks or months, resulting in an unevenly sampled supervisory signal for flow rate estimation. In contrast, auxiliary measurements such as wellhead pressure (WHP), downhole pressure (DHP), temperatures, choke settings, and downstream pressures are available at higher temporal resolution. Nevertheless, these signals may still exhibit missing values, sensor dropouts,

or extended periods of constant readings associated with shut-ins or instrumentation limitations.

The dependent variables considered in this study are the phase flow rates - oil rate (q_o), water rate (q_w), and gas rate (q_g) - primarily obtained from well test measurements. The independent variables comprise pressure, temperature, and operational parameters that are physically linked to well inflow and surface flow behavior.

3.2.1 Independent and Reference Variables

The raw dataset for each well contains a combination of wellhead measurements, downhole measurements, choke parameters, well test rate estimations and MPFM rate estimations. Table 3.1 summarizes the independent operational variables of the dataset, while Table 3.2 lists the geometrical parameters associated with each well.

Although geometrical parameters such as tubing diameter, measured depth, and completion geometry play a fundamental role in governing multiphase flow behavior, they are **not used as independent input variables** in the proposed modeling architecture. This is because the operational measurements - such as pressures, temperatures, and choke settings are themselves physical responses of the well system that inherently embed the effects of the underlying geometry. Consequently, explicitly including geometrical parameters as model inputs would introduce redundancy and may lead to identifiability issues or spurious correlations, particularly under sparse and irregular well test supervision. Instead, geometrical data are used exclusively to provide **initial estimates and constraints during the physics-based model calibration stage**. These parameters inform the initialization of flow equations and serve as fixed structural inputs to the physics model, while the operational variables act as the primary explanatory signals for dynamic rate prediction. This separation ensures physical consistency, reduces model complexity, and improves robustness of the hybrid framework when applied across wells with limited historical data.

Table 3.1
Independent operational variables

Variable	Description	Unit
dhp	Downhole pressure	bara
dht	Downhole temperature	°C
whp	Wellhead pressure	bara
wht	Wellhead temperature	°C
dcp	Downstream choke pressure	bara
choke	Downstream choke regulation	%
well_id	Well unique id	unitless (string)

Table 3.4 summarizes the reference ground truth variables of the dataset. A key modelling decision is the use of **well test measurements** (q_o^{WT} , q_g^{WT} , q_w^{WT}) as the

Table 3.2

Geometrical static data

Variable	Description	Unit
dh_md	Downhole measured depth	m
dh_tvd	Downhole true vertical depth	m
wh_md	Wellhead measured depth	m
wh_tvd	Wellhead true vertical depth	m

Table 3.3

Gas Lift Variables

Variable	Description	Unit
gl_mass_rate	Gas lift injection mass rate	kg/h
gl_open_ratio	Fraction of active gas lift valves	unitless (ratio)

primary ground-truth supervisory targets, rather than MPFM flow rates. This choice is justified as follows:

- Well tests provide **officially calibrated** and **field-accepted** flow-rate measurements, typically validated using test separator systems.
- MPFM measurements, while continuous, can exhibit **bias, drift, salinity sensitivity**, and **phase inversion errors**, especially under varying water-cut and pressure conditions. MPFM estimations are instead used as an industry-standard reference to assess the performance of the proposed model.
- For long-term production modelling and scientific evaluation, well-test values serve as a more reliable **ground-truth reference**.

Table 3.4

Reference ground truth variables

Variable	Description	Unit	Source
qo_well_test	Well test measured oil rate	Sm ³ /h	Well test
qg_well_test	Well test measured gas rate	Sm ³ /h	Well test
qw_well_test	Well test measured water rate	Sm ³ /h	Well test

Table 3.6 summarizes the availability and temporal coverage of historical well test measurements used in this study. A total of 834 well test records are available

Table 3.5
MPFM estimated rates

Variable	Description	Unit	Source
qo_mpfm	MPFM measured oil rate	Sm ³ /h	MPFM
qg_mpfm	MPFM measured gas rate	Sm ³ /h	MPFM
qw_mpfm	MPFM measured water rate	Sm ³ /h	MPFM

across seven production wells, spanning a period from March 2017 to August 2023. The number of records per well varies substantially, reflecting differences in operational practices, testing frequency, and data availability over the field lifecycle. Such sparsity and irregularity are characteristic of well test data, as production tests are typically conducted intermittently due to operational constraints, cost considerations, and production priorities, rather than being continuously acquired like sensor-based measurements. Consequently, the resulting dataset exhibits limited sample sizes and non-uniform temporal spacing, which poses challenges for data-driven modeling approaches. This motivates the adoption of physics-based and hybrid modeling strategies in this work, which are better suited to learning from sparse, irregular, and heterogeneous well test observations.

Table 3.6
Summary of available historical measurements for each production well

Well ID	Number of Records	Start Date	End Date
W06	137	2017-03-06	2023-05-07
W08	69	2018-07-30	2023-06-03
W10	222	2017-03-13	2023-07-31
W11	154	2017-03-01	2023-08-04
W15	86	2017-03-11	2023-05-13
W18	89	2018-05-03	2023-08-09
W19	77	2017-09-02	2023-07-27
All	834	2017-03-01	2023-08-09

Downhole pressure and temperature measurements, shown in Figures 3.2 and 3.3, respectively, represent the thermodynamic conditions at the producing interval. These variables are critical for characterizing reservoir drawdown, inflow behavior, and fluid properties, and they directly influence the physics-based inflow and vertical lift formulations employed in the model. Figures 3.4 and 3.5 illustrate the wellhead pressure and temperature measurements. These measurements capture surface operating conditions and reflect the combined effects of reservoir inflow, multiphase flow in the wellbore, and surface choke control. Wellhead measurements are typically more frequently avail-

able and are therefore particularly valuable for continuous production monitoring and data-driven model calibration. The downstream choke pressure and choke regulation signals are shown in Figures 3.6 and 3.7. These variables describe the imposed flow restriction and surface control strategy applied to the well. Variations in choke opening and downstream pressure directly affect the pressure drawdown across the choke and, consequently, the produced flow rates. As such, these measurements play an important role in capturing operational transients and control-induced production variability.

Collectively, the measurements presented in Figures 3.2–3.7 constitute the primary operational dataset used for model training, calibration, and validation. Their combined use enables the hybrid modeling framework to exploit both physics-based consistency and data-driven adaptability under varying operating conditions.

Figure 3.2
W10 – Downhole pressure

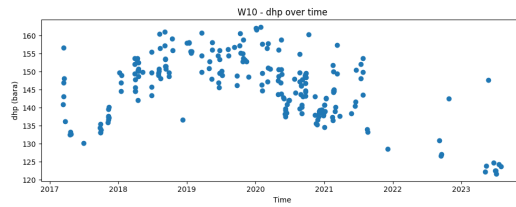


Figure 3.3
W10 – Downhole temperature

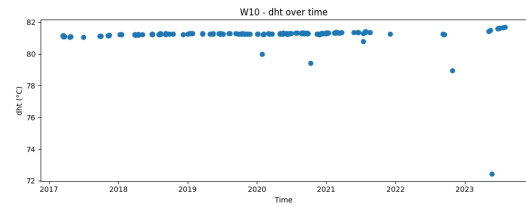


Figure 3.4
W10 – Wellhead pressure

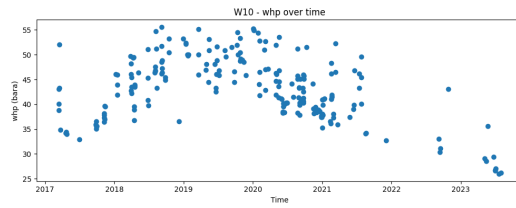


Figure 3.5
W10 – Wellhead temperature

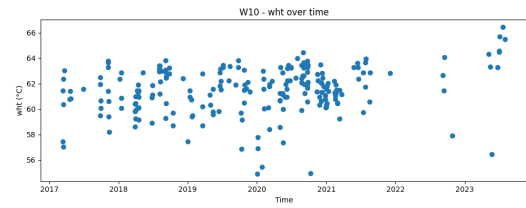


Figure 3.6
W10 – Downstream choke pressure

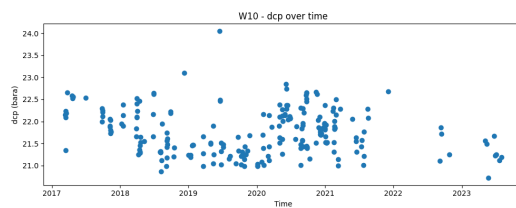
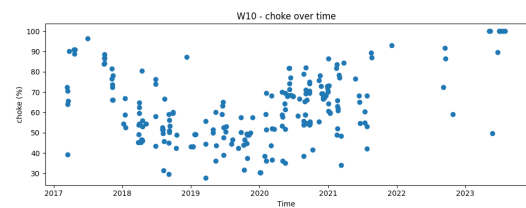


Figure 3.7
W10 – Downstream choke regulation



3.2.2 Well-Level Preprocessing

- **Well-wise data isolation and temporal ordering:** The preprocessing is performed independently for each well by isolating records associated with a single well identifier. Observations are sorted in chronological order to preserve the original temporal structure of the data.
- **Removal of physically invalid sensor readings:** To ensure physical consistency and numerical stability, observations containing zero-valued measurements in key operational variables are removed. These include downhole pressure, wellhead pressure, downstream choke pressure, downhole temperature, wellhead temperature, and choke opening. Zero values are interpreted as sensor dropouts, invalid measurements, or non-operational conditions.
- **Consistent well identification and encoding:** The well identifier is explicitly reassigned to each processed subset to ensure consistency. In addition, a numerical well code is generated from the categorical well identifier to enable machine-learning models to condition on well identity in a numerically stable manner.
- **Normalization of choke opening:** The choke opening is converted from a percentage representation to a normalized fraction by dividing by 100. This transformation ensures numerical consistency and improves convergence behavior in downstream machine-learning components.
- **Removal of incomplete observations:** Observations containing missing values in either dependent variables or independent input variables are removed. This step ensures that all retained samples are fully defined for model training, calibration, and validation.
- **Variable selection and schema enforcement:** After preprocessing, the dataset is restricted to a predefined set of required variables. This enforces a consistent data schema across all wells and prevents unintended variables from entering the modeling pipeline.
- **Enforcement of physically admissible flow rates:** Observations with negative well-test oil, gas, or water rates are excluded to maintain physical realism and prevent numerical instability during model fitting.
- **Treatment of zero well-test water rates:** Reported zero water rates in well-test data may correspond either to genuinely water-free production or to measurements below the detection resolution of the test separator. To mitigate numerical instability and distortion of error metrics, zero water-rate values occurring after the onset of water production for a given well are treated as left-censored

observations and replaced with the minimum non-zero water rate reported for that well. Zero water-rate values prior to the first observed water production are retained unchanged.

- **Exclusion of non-producing operating states:** Rows corresponding to non-producing conditions, identified by simultaneous zero oil and gas well-test rates, are removed. These conditions are not representative of active production behavior and are therefore excluded from the modeling dataset.

3.2.3 Derivation of Multiphase Flow Variables

Additional flow-related variables are derived from the available MPFM measurements solely for the purpose of evaluating MPFM estimates against well-test data, and are not used for physics-model calibration or hybrid model training. The gas-oil ratio (GOR) is computed as:

$$\text{gor_mpfm} = \frac{qg_mpfm}{qo_mpfm}, \quad (3.1)$$

where gor_mpfm and qg_mpfm denote the MPFM measured gas and oil flow rates, respectively.

The water flow rate is reconstructed from the measured water cut using the relationship

$$qw_mpfm = \frac{wc_mpfm \cdot qo_mpfm}{1 - wc_mpfm}, \quad (3.2)$$

where wc is the water cut expressed as a fraction. This formulation ensures consistency between oil, water, and water cut measurements and avoids reliance on potentially noisy direct water rate measurements.

3.2.4 Temporal Index Construction

Following well-level preprocessing, a unified temporal index is constructed to enable consistent time-series modeling across wells. A reference time is defined as the earliest timestamp present in the dataset. For each observation, a discrete time index is computed as

$$\text{time_idx} = \left\lfloor \frac{t - t_{\text{ref}}}{\Delta t} \right\rfloor, \quad (3.3)$$

where t is the observation timestamp, t_{ref} is the reference time, and Δt corresponds to the base time resolution of the data. This integer-valued index provides a compact and model-friendly representation of time while preserving relative temporal ordering.

3.2.5 Cross-Well Consolidation and Encoding

After preprocessing individual wells, the processed datasets are concatenated into a single multi-well dataframe. To support machine-learning models that require numerical inputs, categorical well identifiers are encoded into continuous numerical codes using category-based indexing. This encoding preserves well identity information while remaining compatible with standard regression models.

The final output of the preprocessing stage is a cleaned, physically consistent, and temporally ordered dataset containing all required input variables, derived flow quantities, and auxiliary identifiers required for subsequent physics-based modeling and machine-learning residual learning.

3.3 Physics-Based Inflow Modelling

The physics-based inflow model provides the first-principles foundation for predicting oil, water, and gas flow rates using measurable wellhead and downhole variables (Ahmed, 2010). This model serves as the structured prior within the hybrid learning framework. It maps the instantaneous well state - including downhole pressure, flowing temperature, choke opening, and pressure drop - to a set of physically constrained rate predictions. The model consists of:

1. A pressure-normalized liquid-rate inflow equation that models the total produced liquid rate as a nonlinear function of flowing downhole pressure relative to an estimated reservoir pressure.
2. A logistic water-cut closure that maps choke setting, pressure, and temperature variables to a bounded water fraction, ensuring physically consistent phase partitioning of the total liquid rate.
3. A pressure-driven gas-rate formulation in which gas production is governed by the square-root pressure drawdown and a nonlinear choke effectiveness function, capturing flow control and gas deliverability effects.
4. A parameter calibration procedure based on constrained nonlinear least-squares calibration, jointly fitting liquid, water, and gas rate observations through normalized residual minimization.

3.3.1 Modeling Philosophy

In practical production environments, detailed mechanistic multiphase flow models require extensive fluid property data, well geometry, and calibration effort, which may not be consistently available for all wells (Beggs and Brill, 1973; Ahmed, 2010).

Moreover, such models may still suffer from uncertainty under changing operating conditions. Therefore, a simplified physics-based formulation is adopted, prioritizing robustness, interpretability, and compatibility with sparse supervision. The physics-based model is designed to:

- Capture first-order relationships between pressure, choke setting, and flow rates
- Maintain physical plausibility across operating regimes
- Provide smooth and stable predictions suitable for residual learning
- Allow independent calibration for each well

3.3.2 Model Assumptions and Physical Basis

The model assumes that the flowing downhole pressure, $P_{wf}(t)$, dominantly governs the total liquid production rate from the well. The reservoir is represented using an effective reservoir pressure P_{res} , treated as an unknown parameter to be estimated from the data. The following assumptions are embedded in the formulation:

- The total liquid rate $q_L(t)$ follows a quadratic inflow relationship with respect to the normalized flowing downhole pressure P_{wf}/P_{res} (Vogel, 1968).
- Water cut is determined by operational variables (choke opening, pressure drop, temperatures), using a logistic regression mapping to ensure $0 \leq wc(t) \leq 1$.
- Gas production is driven by a pressure drawdown term proportional to $\sqrt{P_{res} - P_{wf}}$, modulated by a nonlinear choke effectiveness function, representing flow through a restriction under choked or near-choked conditions.
- Oil, water, and gas rates remain non-negative due to clipping operations.

All relationships act instantaneously and do not assume temporal memory, which enables the model to make predictions at any arbitrary timestamp given the set of independent well measurements.

3.3.3 Liquid-Rate Inflow Equation

Total liquid production is modeled using a pressure-driven inflow relationship inspired by classical inflow performance relationships. For wells operating below bubble-point pressure or under multiphase flow conditions, quadratic inflow formulations provide a reasonable approximation of nonlinear pressure-rate behavior while remaining computationally simple. The total liquid rate is expressed as (Vogel, 1968):

$$q_L = q_{L,\max} \left(1 - a \frac{P_{wf}}{P_{res}} - b \left(\frac{P_{wf}}{P_{res}} \right)^2 \right) \quad (3.4)$$

where:

- q_L is the total liquid (oil + water) production rate
- $q_{L,\max}$ represents the theoretical maximum liquid rate
- P_{wf} is the downhole flowing pressure
- P_{res} is an effective reservoir pressure
- a and b are empirical coefficients calibrated per well

This formulation captures the nonlinear decline in liquid production with increasing flowing pressure while remaining flexible enough to represent a wide range of well behaviors. The quadratic term allows the model to accommodate deviations from ideal Darcy flow caused by multiphase effects and near-wellbore pressure losses.

3.3.4 Logistic Water-Cut Closure Model

Phase splitting between oil and water is modeled using a logistic water-cut closure. Logistic functions are widely used in petroleum engineering to represent bounded variables such as water cut due to their smoothness and asymptotic behavior ([Hastie et al., 2009](#)).

Water cut is defined as:

$$wc = wc_{min} + (wc_{max} - wc_{min}) \cdot \sigma(\mathbf{X}_{wc}\mathbf{A}), \quad (3.5)$$

where:

- wc is the water cut,
- $\sigma(\cdot)$ denotes the logistic (sigmoid) function,
- $\mathbf{X}_{wc}$ is a feature vector constructed from operational variables such as choke opening, pressures, temperatures, and interaction terms,
- $\mathbf{A}$ is a vector of learnable coefficients,
- wc_{min} and wc_{max} are lower and upper bounds on water cut.

In this work, $wc_{min} = 0.02$ and $wc_{max} = 0.98$, ensuring that predicted water cut remains within physically plausible limits while avoiding numerical instability near the extremes.

The logistic function can be written explicitly as:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (3.6)$$

Substituting the definition of the logistic function, the water-cut model can be written explicitly as:

$$wc = wc_{min} + (wc_{max} - wc_{min}) \cdot \frac{1}{1 + \exp(-\sum_{i=1}^n X_{wc,i} A_i)} \quad (3.7)$$

The logistic formulation provides a smooth and nonlinear mapping between operational conditions and phase behavior, enabling the model to capture transitions between oil-dominated and water-dominated flow regimes. The inclusion of operational variables allows the model to account for the influence of pressure drawdown, thermal conditions, and choke regulation on phase distribution.

Oil and water production rates are then computed as:

$$q_w = wc \cdot q_L, \quad q_o = (1 - wc) \cdot q_L, \quad (3.8)$$

which enforces consistency with the total liquid rate:

$$q_L = q_o + q_w. \quad (3.9)$$

The parameters $\mathbf{A}$ are estimated during model calibration as part of the overall nonlinear least squares optimization using well-test measurements of oil and water rates. Consequently, water cut is not directly fitted but is implicitly inferred through the coupled phase-splitting relationships.

3.3.5 Gas Flow Modelling

Gas production is modeled using a pressure-driven formulation modulated by choke effectiveness. In producing wells, gas flow is influenced by reservoir pressure drawdown and surface flow restrictions imposed by choke settings. Classical gas deliverability theory indicates that gas flow depends on pressure-squared differences between the reservoir and the wellbore ([Ahmed, 2010](#); [Rawlins and Schellhardt, 1935](#)). Classical gas deliverability theory is often expressed as:

$$q_g = C (P_{res}^2 - P_{wf}^2)^n, \quad (3.10)$$

In contrast, the present work adopts a simplified pressure-drawdown formulation using a square-root dependency. This approximation provides improved numerical stability and computational efficiency while retaining the primary dependence on pressure drawdown. The gas flow rate is expressed as:

$$q_{g,res} = C_g \cdot S_g \cdot \sqrt{P_{res} - P_{wf}} \cdot \sigma(k_{ch}(ch - ch_0)), \quad (3.11)$$

where:

- $q_{g,res}$ is the reservoir gas production rate,
- C_g is a gas flow coefficient,
- S_g is a scaling factor accounting for gas properties,
- P_{res} and P_{wf} are reservoir and bottom-hole pressures, respectively,
- ch denotes the choke opening,
- k_{ch} and ch_0 control choke sensitivity and activation threshold,
- $\sigma(\cdot)$ denotes the logistic (sigmoid) function.

The square-root pressure dependency provides a simplified approximation of pressure-driven gas flow, offering numerical stability and robustness across varying operating conditions. The logistic choke term captures the nonlinear influence of choke opening on gas throughput while ensuring smooth and bounded behavior, thereby avoiding discontinuities associated with piecewise choke models.

When gas lift is present, an additional contribution is included:

$$q_g = q_{g,res} + q_{g,lift}, \quad (3.12)$$

$$q_{g,lift} = C_{gl} \cdot \dot{m}_{gas} \cdot OR, \quad (3.13)$$

where C_{gl} is a gas lift efficiency coefficient, $\dot{m}_{gas}$ is the injected gas mass rate, and OR is the gas lift opening ratio. In the absence of gas lift signals, this term is set to zero.

This formulation provides a computationally efficient and stable approximation of gas flow behavior, while retaining the key dependency on pressure drawdown and operational controls.

3.3.6 Parameter Calibration via Nonlinear Least Squares

All physics model parameters are calibrated independently for each well using nonlinear least-squares optimization against available multiphase flow meter measurements. Calibration is performed jointly across oil, water, and gas rates using normalized residuals to ensure balanced fitting across phases. Physically plausible bounds are imposed on all parameters to prevent non-physical solutions and improve numerical stability.

This per-well calibration strategy allows the model to adapt to well-specific characteristics such as productivity, completion design, and reservoir conditions while maintaining a consistent functional form across wells. The objective of the parameter estimation procedure is to identify physically plausible model parameters that minimize the discrepancy between physics-based flow predictions and available ground-truth flow measurements obtained from multiphase flow meters.

3.3.6.1 Problem Formulation

Let $\boldsymbol{\theta}$ denote the vector of physics model parameters to be estimated for a given well. This parameter vector includes inflow coefficients, water-cut closure parameters, and gas flow coefficients. For a set of N observation times at which measured flow rates are available, the physics model produces predicted oil, water, and gas rates:

$$\hat{\mathbf{q}}_i(\boldsymbol{\theta}) = \begin{bmatrix} \hat{q}_{o,i}(\boldsymbol{\theta}) \\ \hat{q}_{w,i}(\boldsymbol{\theta}) \\ \hat{q}_{g,i}(\boldsymbol{\theta}) \end{bmatrix}, \quad i = 1, \dots, N, \quad (3.14)$$

while the corresponding measured flow rates are denoted by:

$$\mathbf{q}_i = \begin{bmatrix} q_{o,i} \\ q_{w,i} \\ q_{g,i} \end{bmatrix}. \quad (3.15)$$

The residual vector at each observation time is defined as the difference between measured and predicted flow rates:

$$\mathbf{r}_i(\boldsymbol{\theta}) = \mathbf{q}_i - \hat{\mathbf{q}}_i(\boldsymbol{\theta}). \quad (3.16)$$

3.3.6.2 Weighted Least Squares Objective Function

To account for differences in magnitude and measurement uncertainty across phases, a weighted least squares objective function is employed. The total cost function is formulated as:

$$J(\boldsymbol{\theta}) = \sum_{i=1}^N \mathbf{r}_i(\boldsymbol{\theta})^\top \mathbf{W} \mathbf{r}_i(\boldsymbol{\theta}), \quad (3.17)$$

where $\mathbf{W}$ is a diagonal weighting matrix:

$$\mathbf{W} = \begin{bmatrix} w_o & 0 & 0 \\ 0 & w_w & 0 \\ 0 & 0 & w_g \end{bmatrix}. \quad (3.18)$$

The weights w_o , w_w , and w_g are selected to balance the contribution of oil, water, and gas residuals, preventing any single phase from dominating the optimization due to scale differences or higher variability.

3.3.6.3 Nonlinear Least Squares Optimization

The parameter estimation problem is posed as a constrained nonlinear least squares optimization (Levenberg, 1944; Nocedal and Wright, 2006):

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}), \quad (3.19)$$

subject to physically motivated bounds:

$$\boldsymbol{\theta}_{\min} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{\max}. \quad (3.20)$$

All parameters are estimated via nonlinear least-squares calibration against well-test data and represent effective coefficients that compensate for unmodeled multiphase flow, surface interactions, and operational variability. These bounds enforce physical plausibility, such as non-negative productivity indices, reasonable reservoir pressures, and monotonic choke response, and improve numerical stability and convergence behavior.

The optimization is solved using a trust-region-based nonlinear least squares algorithm, which iteratively linearizes the residual function around the current parameter estimate and updates the parameters by solving a sequence of local quadratic subproblems (Nocedal and Wright, 2006). This approach is well-suited for moderately nonlinear models and provides robust convergence under noisy and sparse data conditions. The set of model parameters calibrated during physics-model fitting, together with their physical interpretation and units, is summarized in Table 3.7.

3.3.6.4 Initialization and Regularization

Initial parameter estimates are chosen based on engineering heuristics and prior operational knowledge, such as typical reservoir pressure ranges and expected choke sensitivities. Careful initialization reduces the risk of convergence to non-physical local minima. To mitigate overfitting under sparse measurement conditions, implicit regularization is achieved through:

- Bounded parameter constraints
- Joint calibration across all available phases
- Smooth functional forms in the physics model

Explicit regularization terms are not introduced to preserve the interpretability of calibrated parameters.

3.3.6.5 Handling Sparse and Irregular Measurements

The nonlinear least squares formulation naturally accommodates sparse and irregular measurement schedules, as the objective function is evaluated only at timestamps where flow rate measurements are available. No interpolation or artificial densification of measured flow data is performed during calibration, thereby avoiding bias and preserving the integrity of the observed data.

3.3.6.6 Calibration Quality Assessment

Following optimization, calibration quality is assessed using residual diagnostics and goodness-of-fit metrics computed on the measured data. Visual inspection of predicted versus measured flow rates and residual time series is also performed to identify systematic biases or regime-dependent errors.

Importantly, the calibrated physics-based model is not expected to achieve perfect agreement with measured data. Instead, it is intended to capture dominant flow behavior while leaving systematic discrepancies to be addressed by the subsequent residual learning component.

3.3.6.7 Role in the Hybrid Modeling Architecture

The independently calibrated physics-based model serves as a physically grounded baseline within the hybrid architecture. By constraining the solution space through first-principles relationships and calibrated parameters, the nonlinear least squares estimation stage enables the residual learning model to focus on compensating for unmodeled dynamics, thereby improving overall prediction accuracy, robustness, and data efficiency.

3.4 Machine-Learning Residual Modelling

The physics-based inflow model provides physically consistent baseline predictions for oil, water, and gas production. However, real production behaviour is often influenced by complex wellbore dynamics, operational disturbances, multiphase interactions, and other effects that are not explicitly captured by the simplified analytical physics model. To account for these discrepancies, a machine learning correction stage is introduced. This stage learns the residual structure between observed rates and their corresponding physics-based predictions, producing a hybrid physics - ML model with substantially improved accuracy and generalisation.

The machine-learning residual subsystem operates on a feature set derived from instantaneous measurements and lagged historical values, and predicts additive corrections in logarithmic space for oil, gas, and water flow rates.

3.4.1 Residual Learning Model Philosophy

The use of residual learning instead of direct rate prediction provides several advantages ([He et al., 2016](#)):

- **Preservation of physical structure:** The physics model enforces consistency between pressure, choke settings, phase split, liquid rate, and gas rate. By learning residuals, the machine-learning model augments rather than replaces physically plausible behaviour.
- **Reduction of model complexity:** The machine-learning component learns only deviations from the physics-based predictions, which are typically smoother and of smaller magnitude. This reduces the effective variance of the learned model.
- **Improved generalisation:** Because the residual model does not need to learn fundamental inflow and choke relationships, it exhibits improved generalisation to unseen operating conditions and wells.
- **Enhanced interpretability:** The learned residuals provide direct insight into where and when the physics-based model diverges from observed data, thereby supporting diagnostic analysis and model refinement.
- **Enhanced scalability:** The residual learning formulation supports scalable deployment across large well populations by decoupling well-specific physics calibration from global data-driven learning. Physics models are calibrated independently for each well, while a single machine-learning residual model is shared across all wells. This design notably reduces training and inference overhead, enabling efficient application of virtual flow metering in assets comprising thousands of wells.
- **Improved robustness under sparse and irregular measurements:** The residual learning formulation leverages physics-based predictions as a stable baseline, allowing reliable flow-rate estimation even when ground-truth measurements are sparse, irregular, or intermittently unavailable. By learning residuals only where reference measurements exist and pooling information across multiple wells, the framework reduces sensitivity to data gaps and supports robust operation under realistic field data constraints.

Overall, the machine-learning residual modelling component serves as a complementary layer that enhances physics-based predictions while respecting physical structure and industrial deployment requirements.

3.4.2 Residual Learning Formulation

Let $\hat{q}_o^{\text{phys}}(t)$, $\hat{q}_w^{\text{phys}}(t)$, and $\hat{q}_g^{\text{phys}}(t)$ denote the oil, water, and gas flow rates predicted by the calibrated physics model at time t . The corresponding measured flow rates obtained from multiphase flow meters are denoted by $q_o(t)$, $q_w(t)$, and $q_g(t)$.

Residual learning is performed in logarithmic space to stabilize variance and ensure positivity of predicted flow rates (Box and Cox, 1964). The residual targets are defined as

$$r_i(t) = \log(1 + q_i(t)) - \log\left(1 + \hat{q}_i^{\text{phys}}(t)\right), \quad i \in \{o, w, g\}. \quad (3.21)$$

This formulation allows the residual model to learn multiplicative corrections in the original flow-rate domain while remaining numerically stable for low-rate conditions.

3.4.3 Feature Construction

The input feature vector for residual learning is constructed by concatenating measured operating variables with physics-based flow predictions. For each valid time index t , the feature vector is defined as

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{u}(t) \\ \hat{q}_o^{\text{phys}}(t) \\ \hat{q}_w^{\text{phys}}(t) \\ \hat{q}_g^{\text{phys}}(t) \end{bmatrix}, \quad (3.22)$$

where $\mathbf{u}(t)$ represents the set of independent well measurements, including pressures, temperatures, choke settings, and lagged variables where applicable.

Including physics-based predictions as explicit features enables the residual model to condition its corrections on physically meaningful state estimates, thereby reducing extrapolation risk and improving generalization.

3.4.4 Data Normalization and Polynomial Expansion

Prior to training, the feature matrix is standardized using zero-mean and unit-variance scaling (Hastie et al., 2009),

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (3.23)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are computed from the training dataset.

To capture low-order nonlinear interactions between features, a polynomial feature expansion of degree d is applied,

$$\mathbf{z} = \Phi_d(\tilde{\mathbf{x}}), \quad (3.24)$$

where $\Phi_d(\cdot)$ denotes the polynomial feature mapping without bias terms. In the current implementation, a linear expansion ($d = 1$) is used to maintain computational efficiency and avoid overfitting under sparse data conditions.

3.4.5 Residual Model Architecture

The residual mapping from feature space to residual corrections is learned using a multi-output regression framework,

$$\mathbf{r}(t) = f_{\text{ML}}(\mathbf{z}(t)), \quad (3.25)$$

where $\mathbf{r}(t) = [r_o(t), r_w(t), r_g(t)]^\top$.

The function $f_{\text{ML}}(\cdot)$ is implemented as a *MultiOutputRegressor* with gradient-boosted decision trees as base learners. Specifically, a histogram-based gradient boosting regressor is employed due to its robustness, fast training, and suitability for tabular industrial data (Friedman, 2001). Key hyperparameters are selected to balance accuracy and computational efficiency, including shallow tree depth, early stopping, and minimum leaf size constraints.

A single global residual model is trained using data pooled from all wells, while physics models are calibrated independently for each well. This design enables cross-well learning of common residual patterns while preserving well-specific physical behavior.

3.4.6 Hybrid Prediction and Residual Application

During inference, physics-based predictions are first generated for each well. The trained residual model then produces residual estimates $\hat{r}_i(t)$, which are applied selectively depending on the flow phase and operating regime.

For oil and gas, residual corrections are applied in logarithmic space as

$$\hat{q}_i(t) = \exp\left(\log\left(1 + \hat{q}_i^{\text{phys}}(t)\right) + \hat{r}_i(t)\right) - 1, \quad i \in \{o, g\}. \quad (3.26)$$

For water, residual corrections are applied only when the physics-based water rate exceeds a predefined threshold $q_w^{\min}$, ensuring that spurious corrections are not intro-

duced during dry or near-dry conditions,

$$\hat{q}_w(t) = \begin{cases} \exp(\log(1 + \hat{q}_w^{\text{phys}}(t)) + \hat{r}_w(t)) - 1, & \hat{q}_w^{\text{phys}}(t) > q_w^{\min}, \\ 0, & \text{otherwise.} \end{cases} \quad (3.27)$$

Additional gating rules based on water cut and residual magnitude are applied to oil predictions to prevent physically implausible corrections under extreme conditions.

3.4.7 Consistency Constraints

To preserve physical consistency, predicted oil and water rates are constrained such that

$$\hat{q}_o(t) + \hat{q}_w(t) = \hat{q}_L(t), \quad (3.28)$$

where $\hat{q}_L(t)$ denotes the total liquid rate. Water predictions are clipped to ensure non-negativity and boundedness, and oil rates are adjusted accordingly.

3.5 Hybrid Physics-ML Rate Synthesis

The hybrid physics-machine learning (ML) rate synthesis component defines how physics-based predictions and machine-learning residual corrections are combined to produce the final estimates of oil, water, and gas flow rates. This stage constitutes the final inference step of the proposed architecture and is critical for ensuring physical consistency, robustness, and safe deployment under sparse and irregular measurement conditions.

3.5.1 Sequential Inference Structure

The proposed framework follows a strictly sequential inference strategy. For each well and time index t , physics-based predictions are generated first, followed by selective application of machine-learning residual corrections. Let

$$\hat{q}_o^P(t), \hat{q}_w^P(t), \hat{q}_g^P(t) \quad (3.29)$$

denote the oil, water, and gas flow rates predicted by the calibrated physics model, and let

$$\hat{r}_o(t), \hat{r}_w(t), \hat{r}_g(t) \quad (3.30)$$

denote the corresponding residuals predicted by the machine-learning model.

The physics model is always evaluated regardless of data availability, while the machine-learning component operates as a corrective layer whose influence is controlled through physically motivated gating mechanisms.

3.5.2 Log-Space Residual Application

Residuals are learned and applied in logarithmic space to ensure numerical stability and preserve non-negativity of flow rates. For a given phase $i \in \{o, w, g\}$, the corrected rate is computed as

$$\hat{q}_i(t) = \exp(\log(1 + \hat{q}_i^P(t)) + \hat{r}_i(t)) - 1. \quad (3.31)$$

This formulation corresponds to a multiplicative correction (Box and Cox, 1964) in the original rate domain and reflects the assumption that modeling errors scale with the magnitude of the predicted flow rate.

3.5.3 Phase-Specific Synthesis Rules

The hybrid synthesis logic differs across fluid phases to reflect their distinct flow characteristics and data reliability.

3.5.3.1 Oil Rate Synthesis

Oil rate corrections are applied conditionally. The physics-based oil prediction $\hat{q}_o^P(t)$ is retained unless one or more gating conditions are satisfied. These conditions include elevated water cut or large predicted residual magnitude. When gating is active, the corrected oil rate is computed using log-space residual application:

$$\hat{q}_o(t) = \exp(\log(1 + \hat{q}_o^P(t)) + \hat{r}_o(t)) - 1. \quad (3.32)$$

If gating conditions are not satisfied, the physics-based oil rate is preserved.

3.5.3.2 Gas Rate Synthesis

Gas rates are corrected unconditionally using the residual model, reflecting the strong dependence of gas flow on pressure drawdown and choke behavior:

$$\hat{q}_g(t) = \exp(\log(1 + \hat{q}_g^P(t)) + \hat{r}_g(t)) - 1. \quad (3.33)$$

The corrected gas rate is subsequently clipped to ensure non-negativity.

3.5.3.3 Water Rate Synthesis

Water rate corrections are applied only when the physics-based water rate exceeds a predefined minimum threshold $q_w^{\min}$:

$$\hat{q}_w(t) = \begin{cases} \exp(\log(1 + \hat{q}_w^P(t)) + \hat{r}_w(t)) - 1, & \hat{q}_w^P(t) > q_w^{\min}, \\ 0, & \text{otherwise.} \end{cases} \quad (3.34)$$

This gating strategy prevents spurious corrections during dry or near-dry conditions where measurement noise and model uncertainty are high.

3.5.4 Physical Consistency Enforcement

Following residual application, additional consistency constraints are enforced to ensure physically meaningful predictions. The total liquid rate is defined as

$$\hat{q}_L(t) = \hat{q}_o(t) + \hat{q}_w(t). \quad (3.35)$$

Water rates are clipped to ensure

$$0 \leq \hat{q}_w(t) \leq \hat{q}_L(t), \quad (3.36)$$

and the oil rate is adjusted accordingly to preserve liquid consistency. All phase rates are constrained to be non-negative.

3.5.5 Role of Gating and Safety Mechanisms

The synthesis stage incorporates multiple gating and clipping mechanisms to limit machine-learning influence in poorly observed regimes. These safeguards reduce extrapolation risk under sparse or irregular measurements and ensure that physics-based predictions dominate when data-driven corrections are unreliable.

By design, the hybrid synthesis logic prioritizes physical plausibility, robustness, and stable behavior over unconstrained accuracy improvements.

3.5.6 Summary of Hybrid Rate Assembly

The hybrid physics-ML rate synthesis combines deterministic physics-based predictions with selectively applied machine-learning residual corrections. This structured integration enables improved predictive accuracy while preserving interpretability, physical consistency, and scalability across large well populations.

3.6 Model Integration

3.6.1 Step-by-Step Pipeline Summary

The overall modeling workflow of the proposed hybrid framework consists of a sequence of structured steps that integrate physics-based modeling with data-driven residual learning. The pipeline is designed to ensure causal consistency, scalability across wells, and robustness under both temporal and spatial generalization settings.

1. **Data preparation and preprocessing:** Raw well data, including pressure, temperature, choke settings, and well-test measurements, are first cleaned and filtered to remove invalid or missing entries. Relevant features are normalized and aligned temporally. Well-test rates are treated as the reference ground truth for model training and evaluation.
2. **Temporal data partitioning:** For each well, the dataset is split into training, validation, and test sets using a lag-safe temporal splitting strategy. This ensures that future information is not leaked into the training phase and preserves causal consistency. For spatial generalization, a LOWO strategy is employed, where one well (e.g., W10) is held out entirely for testing.
3. **Physics model calibration:** A physics-based multiphase flow model is calibrated independently for each well using nonlinear least squares optimization. The calibration process estimates well-specific parameters such as reservoir pressure and flow coefficients, enabling the model to capture fundamental physical behavior.
4. **Physics-based prediction:** Using the calibrated parameters, the physics model is used to generate baseline predictions for oil, gas, and water flow rates across all samples. These predictions serve as the foundation for subsequent residual learning.
5. **Residual computation:** Residuals are computed as the difference between observed well-test measurements and physics-based predictions. These residuals represent systematic discrepancies arising from unmodeled physical effects, measurement noise, or simplifying assumptions in the physics model.
6. **Feature engineering for residual learning:** A reduced feature set is constructed using selected operational variables (e.g., pressure, choke) and physics-derived quantities. Additional transformations, such as log-space representation and regime-based grouping, are applied to improve learning stability and capture nonlinear relationships.

7. **Residual model training:** A machine learning model is trained globally across multiple wells to predict residual corrections. The use of pooled data enables the model to learn shared patterns in physics deviations, improving generalization across wells.
8. **Hybrid prediction:** Final predictions are obtained by combining physics-based estimates with the predicted residual corrections. Gating mechanisms and physical constraints are applied to ensure that corrected predictions remain within physically plausible limits.
9. **Model evaluation:** The hybrid model is evaluated using well-test measurements as the reference ground truth. Performance is assessed under both temporal and spatial generalization settings using metrics such as R^2 , MAE, RMSE, and MAPE. Additionally, model performance is compared against MPFM estimations, which serve as an industry-standard benchmark.
10. **Model persistence:** Calibrated physics parameters and trained residual models are stored to enable reuse across wells and deployment scenarios. This avoids the need for repeated full retraining and ensures consistency in model application.
11. **Deployment:** For a new or unseen well, the physics-based model is calibrated using a limited set of well-test measurements to capture well-specific characteristics. The globally trained residual model is then applied without retraining, enabling efficient commissioning with minimal data requirements.
12. **Operation:** During operation, the hybrid model generates flow rate predictions at discrete time intervals using available operational inputs such as pressure, temperature, and choke settings. This supports near real-time production monitoring in practical field environments.

3.6.2 Computational Considerations

The proposed hybrid modeling architecture combines a physics-based flow model with a data-driven residual learning component. While this integration improves predictive accuracy and generalization under sparse data conditions, it introduces several computational considerations related to efficiency, scalability, numerical stability, and deployment feasibility. These aspects are discussed below.

3.6.2.1 Model Decomposition and Computational Load

The hybrid architecture is deliberately decomposed into two stages: (i) a physics-based forward model and (ii) a machine learning model trained to predict residual errors. This decomposition allows the majority of physical computations to remain lightweight,

relying on closed-form or low-dimensional parametric equations. Consequently, the computational complexity of the physics layer scales linearly with the number of samples, i.e.,

$$\mathcal{O}(N), \quad (3.37)$$

where N denotes the number of time steps.

The machine learning residual model operates on a reduced feature space, typically consisting of selected operational variables and physics-derived quantities. As a result, the hybrid model avoids the high-dimensional input spaces commonly encountered in end-to-end deep learning approaches, thereby reducing both training time and memory requirements.

3.6.2.2 Parameter Calibration and Training Efficiency

Parameter calibration in the physics layer is performed using nonlinear least squares optimization. Given the low dimensionality of the parameter vector, convergence is typically achieved within a small number of iterations. The computational cost of this calibration step is therefore modest and well-suited for per-well or batched multi-well calibration.

The residual learning component is trained after physics model calibration, which decouples physical parameter estimation from machine learning optimization. This sequential training strategy improves numerical stability and reduces the risk of ill-conditioned joint optimization. Furthermore, it allows reuse of calibrated physics parameters when transferring the model to new wells with limited data.

3.6.2.3 Scalability Across Multiple Wells

From a scalability perspective, the hybrid framework supports both per-well and global learning paradigms. Physics model calibration can be executed independently for each well, enabling embarrassingly parallel computation across wells. The residual model, on the other hand, may be trained globally using pooled data to capture cross-well patterns, with computational complexity depending on the chosen learning algorithm.

This modular design allows the framework to scale efficiently with the number of wells while maintaining manageable computational overhead, making it suitable for field-scale virtual flow metering applications.

3.6.2.4 Numerical Stability and Robustness

The inclusion of physics-based constraints inherently improves numerical stability by restricting predictions to physically plausible regimes. Additionally, normalization of inputs and careful handling of unit consistency (e.g., pressure in bar absolute and tem-

perature in degrees Celsius) are employed to prevent numerical scaling issues during training.

Gating mechanisms used to combine physics predictions and residual corrections further enhance robustness by attenuating machine learning corrections under extrapolative or low-confidence conditions. This design reduces sensitivity to noisy measurements and mitigates the risk of unphysical predictions.

3.6.2.5 Inference Time and Deployment Considerations

At inference time, the hybrid model requires a single forward evaluation of the physics equations followed by a lightweight residual prediction. This results in low-latency predictions suitable for near real-time applications. Compared to purely data-driven deep learning models, the proposed approach offers considerably reduced inference cost, making it amenable to deployment in production monitoring systems with limited computational resources.

Overall, the computational design of the hybrid model achieves a favorable balance between accuracy, interpretability, and efficiency, enabling practical deployment in real-world well surveillance and production optimization workflows.

Table 3.7

Parameters and initial guesses used for nonlinear calibration of the physics-based well flow model

Parameter	Initial guess	Unit	Physical interpretation
P_{res}	$P_{\text{wf,max}} + \Delta P$	bara	Effective reservoir pressure inferred from flowing pressure with hydrostatic or heuristic offset
$q_{L,\text{max}}$	$\overline{(q_o + q_w)}$	Sm ³ /h	Maximum liquid production capacity (lumped inflow strength)
a	0.2	–	Linear pressure sensitivity coefficient in the liquid inflow relationship
b	0.5	–	Quadratic pressure sensitivity coefficient accounting for nonlinearity
C_g	50.0	Sm ³ /h/ $\sqrt{\text{bar}}$	Reservoir gas productivity coefficient in the pressure-driven gas-rate formulation
k_{ch}	5.0	–	Choke sensitivity parameter controlling the sharpness of the logistic choke response
ch_0	0.3	–	Reference (midpoint) choke opening for transition from restricted to open flow
C_{gl}	0.01	–	Gas-lift efficiency coefficient scaling injected gas contribution
$A_{\text{wc},1\dots 8}$	0.0	–	Initial coefficients of the logistic water-cut closure model

CHAPTER 4

EXPERIMENTATION

This chapter describes the experimental design, data partitioning strategy, and evaluation protocols used to assess the performance of the proposed physics-informed residual learning virtual flow metering (VFM) model. The experiments are designed to provide a rigorous and realistic assessment of model accuracy, robustness, and generalization capability under conditions that closely resemble practical deployment in oil and gas production environments.

4.1 Experimental Objectives

The primary objectives of the experimental evaluation are to:

1. Quantitatively evaluate the accuracy of oil, water, and gas flow rate predictions produced by the proposed hybrid physics-machine learning model using well-test measurements as reference data. Well test rates are treated as the primary source of truth (gold standard), as they provide the most reliable direct measurements of production rates, although they may still contain operational uncertainties. Assess temporal generalization by evaluating model performance on unseen time periods.
2. Quantify the performance gains achieved through hybrid residual correction model relative to physics-only model predictions across multiple wells.
3. Compare the predictive performance of the proposed modeling approaches against MPFM, in order to assess the added value of the data-driven framework relative to the industry standard MPFM benchmark.
4. Assess the spatial generalization capability of the proposed framework under realistic deployment conditions by evaluating model performance on previously unseen wells using a LOWO strategy. For unseen wells, limited physics-model calibration is permitted, while the global machine-learning residual model is applied without retraining.
5. Benchmark the predictive performance of the proposed hybrid framework against the results reported by (Nemoto et al., 2023), leveraging the fact that both studies are based on the wells from the same oil and gas field. This comparison aims to contextualize the performance of the proposed approach relative to existing cloud-based virtual flow metering methodologies under similar data.

4.2 Production-Mimicking Experimental Workflow

All experiments are conducted using a production-mimicking workflow. Physics models calibrated during the training phase remain fixed throughout the experimentation, and the globally trained machine-learning residual model is applied strictly in inference mode. No retraining or parameter updates are performed during testing.

Let t denote the prediction time index. At each time step t , only measurements available up to that instant are used for prediction, thereby emulating real-time or near-real-time virtual flow metering operation.

4.3 Experimental Model Configurations

Two experimental model configurations are considered:

1. **Physics-only configuration:** Flow rates are estimated using a physics-based model calibrated on a per-well basis. For unseen wells, limited calibration of well-specific physics parameters is performed, while no data-driven correction is applied.
2. **Physics-informed hybrid configuration:** Physics-based predictions are corrected using a globally trained machine-learning residual model operating in log-space, with regime-aware gating and physical constraints enforced. For unseen wells, only the physics-model parameters are calibrated, and the residual model is applied without retraining.

Comparing these configurations allows the incremental contribution of residual learning to be quantified. In addition, the performance of both experimental configurations is evaluated against the multiphase flow meter (MPFM) measurements, which serve as a strong industry baseline for production rate estimation.

4.4 Evaluation Metrics

Model performance is quantified using a set of complementary metrics that capture absolute error, relative error, systematic bias, and goodness of fit. Let y_i denote the measured value of a given flow rate at time index i , $\hat{y}_i$ the corresponding model prediction, and N the number of evaluated samples.

4.4.1 Coefficient of Determination

The coefficient of determination (R^2) is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (4.1)$$

where $\bar{y}$ denotes the mean of the measured values. The R^2 metric is unitless and indicates the proportion of variance in the measured data explained by the model.

4.4.2 Mean Absolute Error

The mean absolute error (MAE) is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (4.2)$$

MAE is expressed in the same physical units as the predicted variable (e.g., Sm^3/h) and provides an intuitive measure of the average magnitude of prediction error.

4.4.3 Root Mean Squared Error

The root mean squared error (RMSE) is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}. \quad (4.3)$$

RMSE is expressed in the same physical units as the predicted variable (e.g., Sm^3/h). Compared to MAE, RMSE penalizes larger errors more strongly and is therefore more sensitive to outliers.

4.4.4 Mean Absolute Percentage Error

The mean absolute percentage error (MAPE), also referred to as the mean absolute relative error (MARE), is defined as:

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (4.4)$$

MAPE is a dimensionless metric reported as a percentage (%). It quantifies the average relative magnitude of prediction error. For low-rate regimes, particularly for water production, MAPE may exhibit inflated values and is therefore interpreted with caution.

4.4.5 Mean Percentage Error

The mean percentage error (MPE), also referred to as the signed mean relative error or average discrepancy, is defined as:

$$\text{MPE} = \frac{100}{N} \sum_{i=1}^N \frac{\hat{y}_i - y_i}{y_i}. \quad (4.5)$$

MPE is a dimensionless metric reported as a percentage (%). It captures systematic prediction bias, with positive values indicating overprediction and negative values indicating underprediction.

4.5 Evaluation of Derived Production Ratios

In addition to individual phase flow rates, the experiments evaluate derived production ratios, including the water-gas ratio (WGR) and gas-oil ratio (GOR), defined as:

$$\text{wgr} = \frac{q_w}{q_g}, \quad \text{gor} = \frac{q_g}{q_o}, \quad (4.6)$$

where q_o , q_w , and q_g denote oil, water, and gas flow rates, respectively.

These ratios are computed from both measured and predicted flow rates, and the same evaluation metrics are applied. Data points associated with very low denominator values are excluded to avoid numerical instability.

4.6 Cross-Well Temporal Generalization Evaluation Setup

The cross-well temporal generalization evaluation is designed to assess the ability of the proposed modeling framework to produce accurate flow-rate predictions on unseen time periods while remaining within the same set of wells. This evaluation reflects a common production monitoring scenario in which historical data are available for a well, and the objective is to predict flow rates under future operating conditions without retraining the model.

For each well, the available time-series data are partitioned into training and testing subsets using a temporally ordered split. Earlier observations are used for physics-model calibration and, where applicable, training of the global machine-learning residual model, while later observations are held out exclusively for evaluation. This strategy preserves the natural temporal ordering of the data and prevents information leakage from future operating conditions into the training process.

The physics-only and hybrid model configurations are evaluated under identical temporal splits. Physics-based model parameters are calibrated using the training portion of the data for each well. The machine-learning residual model is trained globally

using training data from all wells and is applied during inference without retraining. Multiphase flow meter MPFM estimates are evaluated on the same test intervals for fair comparison.

Model performance is quantified using well-test measurements as the reference standard and evaluated using the evaluation metrics defined in Section 4.4 for oil, water, and gas flow rates. Reporting results on a per-well basis enables systematic assessment of temporal robustness under evolving operating conditions.

4.6.1 Experimental Data Partitioning

To ensure robust model development and unbiased performance evaluation, the dataset is partitioned into training, validation, and test subsets on a per-well basis using a temporal hold-out strategy.

For each well, observations are first sorted chronologically, and the data are then split into three contiguous segments according to predefined proportions:

- **Training set:** 70
- **Validation set:** 15
- **Test set:** 15

To further mitigate temporal leakage and autocorrelation effects between subsets, a fixed gap is introduced between consecutive splits. Specifically, a small number of samples separating the training–validation and validation–test boundaries are excluded from all subsets. This ensures that short-term temporal dependencies do not artificially inflate model performance.

Unlike random sampling approaches, this method strictly preserves the temporal ordering of observations, thereby reflecting real-world deployment conditions where models are trained on historical data and evaluated on future unseen data.

The partitioning is performed independently for each well to ensure that: (i) all wells are represented across the training, validation, and test sets, and (ii) well-specific temporal dynamics are preserved without inter-well data leakage.

Wells with insufficient data to satisfy the minimum split requirements (including the gap constraints) are excluded from the analysis to maintain the integrity of the evaluation.

The final datasets are obtained by concatenating the corresponding subsets across all wells. This strategy provides a realistic and physically consistent evaluation framework for time-dependent production data, while avoiding information leakage between training and evaluation phases.

4.7 Spatial Generalization Evaluation Setup

The spatial generalization evaluation assesses the ability of the proposed modeling framework to transfer predictive capability to a previously unseen well. This evaluation is conducted using a targeted Leave-One-Well-Out (LOWO) strategy, designed to emulate a realistic deployment scenario in which a trained model is applied to a new well with limited or no prior exposure during training.

In this setup, the well with the largest number of historical observations (W10) is excluded entirely from model development and used exclusively for evaluation, while the remaining wells constitute the training set. The machine-learning residual model is trained globally using data from the training wells and is kept fixed during evaluation on the unseen well. For the held-out well, limited calibration of well-specific physics parameters is performed using a small subset of available data to ensure physical consistency, while retraining of the residual model is intentionally avoided.

This experimental design serves as a conservative stress test of spatial generalization. By excluding the most data-rich well from training, the evaluation prevents dominance of a single well during residual learning and ensures that model performance on W10 reflects genuine cross-well transferability rather than implicit memorization driven by data availability. The setup therefore provides a stringent assessment of whether the proposed hybrid framework can generalize to a well with complex and well-sampled production behavior based solely on information learned from other wells.

Both physics-only and hybrid model configurations are evaluated on the held-out well, and their performance is compared against MPFM estimates using well-test measurements as the reference standard. Model performance is quantified using the evaluation metrics defined in Section 4.4 for oil, water, and gas flow rates.

4.8 Summary of Experimental Design

The experimental design emphasizes realism, reproducibility, and practical relevance. By enforcing temporal separation, preventing data leakage, and mimicking production deployment conditions, the experiments provide a comprehensive and unbiased assessment of the proposed virtual flow metering model. The results presented in the following chapter build upon this experimental framework to demonstrate the effectiveness of physics-informed residual learning for multiphase flow rate estimation in oil and gas production wells.

Table 4.1
Performance metrics used for model evaluation

Metric	Description	Units
R^2	Coefficient of determination measuring goodness of fit	unitless
MAE	Mean Absolute Error, average magnitude of prediction error	Same as target (e.g., Sm ³ /h)
RMSE	Root Mean Squared Error, penalizing large deviations	Same as target (e.g., Sm ³ /h)
MAPE / MARE	Mean absolute relative prediction error	%
MPE / MRE	Signed mean relative error (average discrepancy)	%

CHAPTER 5

RESULTS AND ANALYSIS

This chapter presents the experimental results and analysis of the proposed hybrid physics-machine learning virtual flow metering framework.

5.1 Temporal Generalization Performance

Temporal generalization evaluates the ability of the proposed model to predict future production behavior using historical data from the same well. This is a critical requirement for virtual flow metering applications, where models must remain reliable under evolving operating conditions, including changes in pressure, choke settings, and multiphase flow regimes.

In this study, temporal generalization is assessed by training the models on earlier time periods and evaluating them on unseen future data for each well. This setup ensures that no future information is used during training, thereby mimicking real-world deployment conditions. The evaluation is conducted across multiple wells with varying flow characteristics, enabling a comprehensive assessment of model robustness and adaptability.

The performance of three approaches is compared: the physics-based model, the proposed hybrid physics-machine learning model, and the MPFM baseline. Metrics including R^2 , MAE, RMSE, and MAPE are used to quantify predictive accuracy and reliability across wells.

5.1.1 Gas Rate (q_g)

Table 5.1 reports gas rate prediction performance under lag-safe temporal extrapolation.

The physics-based model exhibits highly variable performance for gas rate prediction across wells. While moderate accuracy is achieved for wells such as W06 ($R^2 = 0.77$) and W11 ($R^2 = 0.59$), the model fails significantly for several others, including W08, W15, and W19, where strongly negative R^2 values are observed. These results indicate that the physics formulation struggles to capture the complex gas flow behavior under varying operating conditions, particularly in wells with dynamic pressure drawdown and multiphase interactions. The consistently high MAE and RMSE values further highlight the limited predictive capability of the standalone physics model for temporal extrapolation.

In contrast, the hybrid model demonstrates consistently strong performance across all wells, as shown in Table 5.1. High predictive accuracy is achieved for most wells,

with R^2 values exceeding 0.85 and reaching near-perfect levels for W06, W11, and W18. The substantial reduction in MAE and RMSE compared to the physics model confirms the effectiveness of the residual learning framework in correcting systematic biases in gas rate predictions. Notably, even for challenging wells such as W08, where the physics model fails ($R^2 = -30.24$), the hybrid model achieves a positive R^2 of 0.44, demonstrating its ability to generalize across difficult operating regimes. The consistently low MAPE values further indicate stable and reliable predictive performance.

The MPFM baseline generally provides strong performance for gas rate prediction, achieving high R^2 values for wells such as W06, W10, W11, and W15. However, significant degradation is observed for certain wells, particularly W08 and W18, where negative R^2 values and large prediction errors are recorded. This suggests that MPFM measurements are sensitive to flow conditions and may become unreliable under complex multiphase regimes or adverse operating conditions.

Overall, the hybrid model achieves the best balance between accuracy and robustness. It consistently outperforms the physics-based model and demonstrates more stable performance than the MPFM baseline across wells. These results confirm that integrating physics-based modeling with machine learning residual correction significantly enhances temporal generalization capability for gas rate prediction.

Table 5.1
Temporal generalization performance for gas rate (q_g)

Physics Model							
Metric	W06	W08	W10	W11	W15	W18	W19
R^2	0.77	-30.24	-0.06	0.59	-7.23	-0.10	-23.74
MAE	1823.01	3186.58	3163.45	2310.10	3115.71	6685.97	6520.88
RMSE	2204.70	3227.35	3867.35	2937.40	4070.56	7147.10	6579.77
MAPE (%)	13.67	78.41	24.47	21.01	50.32	55.01	89.95
Hybrid Model							
R^2	0.99	0.44	0.97	0.99	0.79	0.99	0.85
MAE	96.74	374.89	447.92	385.44	402.05	614.78	540.34
RMSE	119.94	378.78	640.71	444.37	686.43	749.10	549.49
MAPE (%)	0.95	9.62	3.21	5.83	6.77	4.34	7.53
MPFM							
R^2	0.99	-12.95	0.92	0.93	0.96	-1.71	0.94
MAE	330.88	1699.02	877.14	809.32	252.92	9630.87	311.74
RMSE	366.27	1890.10	1078.73	1244.23	299.93	11138.26	343.36
MAPE (%)	2.38	45.25	6.12	13.63	4.64	46.58	4.07

5.1.2 Oil Rate (q_o)

Table 5.2 summarizes the per-well temporal generalization performance for oil rate prediction.

The physics-based model exhibits highly inconsistent performance across wells. While it achieves moderate accuracy for well W06 ($R^2 = 0.65$), it fails significantly for most other wells, with strongly negative R^2 values (e.g., W08, W11, W15, W19). This indicates that the physics formulation alone is unable to capture the complex and evolving dynamics of multiphase flow under changing operating conditions. The large errors in MAE and RMSE further confirm the limited adaptability of the physics model to temporal variations.

In contrast, the proposed hybrid model demonstrates consistently strong performance across the majority of wells. Near-perfect predictive accuracy is observed for several wells, including W06, W10, W11, and W18, with R^2 values close to 1.0 and very low MAE and RMSE. This highlights the effectiveness of residual learning in correcting systematic biases in the physics-based predictions. Even for more challenging wells such as W15 and W19, the hybrid model maintains reasonable performance, significantly outperforming the physics model.

However, performance degradation is observed for well W08 ($R^2 = -1.87$), indicating that the hybrid model may struggle under certain operating regimes not sufficiently represented in the training data. This suggests that the effectiveness of the residual learning component is dependent on the coverage of flow conditions and regimes seen during training.

The MPFM baseline generally provides strong performance, particularly for wells such as W06, W10, and W15. However, it exhibits instability for certain wells, notably W08 and W18, where negative R^2 values and high errors are observed. This highlights the sensitivity of MPFM measurements to flow conditions and potential measurement uncertainties.

Overall, the hybrid model achieves the best balance between accuracy and robustness. It consistently outperforms the physics-based model and demonstrates competitive, and in many cases superior, performance compared to MPFM. These results confirm that integrating physics-based modeling with machine learning residual correction considerably enhances temporal generalization capability for oil rate prediction.

5.1.3 Water Rate (q_w)

Table 5.3 reports water rate prediction performance under lag-safe temporal extrapolation.

The physics-based model performs poorly for water rate prediction across all wells, with consistently negative R^2 values. This indicates that the model fails to capture the underlying dynamics of water production, particularly under multiphase flow condi-

Table 5.2
Temporal generalization performance for oil rate (q_o)

Physics Model							
Metric	W06	W08	W10	W11	W15	W18	W19
R^2	0.65	-11.39	-2.28	-10.76	-18.77	-8.86	-32.68
MAE	18.53	9.74	47.63	81.93	38.66	105.13	58.87
RMSE	22.63	11.84	67.75	139.10	41.75	112.39	62.59
MAPE (%)	25.09	86.45	61.31	108.27	138.13	217.45	173.15
Hybrid Model							
R^2	0.99	-1.87	0.99	0.99	0.73	0.98	0.88
MAE	0.49	2.42	2.46	3.51	3.46	4.43	3.71
RMSE	0.67	2.43	3.85	4.46	4.65	4.93	3.94
MAPE (%)	0.61	22.04	2.69	5.01	11.46	7.89	11.40
MPFM							
R^2	0.99	-7.08	0.99	0.85	0.95	-0.72	0.86
MAE	1.45	3.84	2.56	8.71	1.84	44.54	2.85
RMSE	1.89	4.07	3.32	16.10	2.08	47.53	4.27
MAPE (%)	1.16	35.45	2.84	20.15	7.76	61.38	6.86

tions where water behavior is highly nonlinear and intermittent. Although the absolute errors (MAE and RMSE) are relatively moderate for some wells, the persistently high MAPE values often approaching or exceeding 100% - highlight the difficulty of accurately modeling water rates using physics alone. These results suggest that the simplified physics formulation is insufficient to represent complex water production mechanisms such as phase interactions, breakthrough events, and low-rate regimes.

In contrast, the hybrid model demonstrates substantial improvement in water rate prediction for most wells, as shown in Table 5.3. Strong performance is observed for wells such as W06, W10, W11, and W15, where positive R^2 values and significantly reduced MAE and RMSE indicate accurate and stable predictions. The hybrid model effectively leverages residual learning to correct systematic biases in the physics model, resulting in consistent performance gains across varying operating conditions. However, reduced performance is observed for wells such as W08 and W18, where R^2 remains low or slightly negative. This suggests that water prediction remains challenging in certain regimes, particularly where water production is highly sparse or exhibits abrupt transitions.

The MPFM baseline shows mixed performance for water rate prediction. While it achieves high accuracy for certain wells (e.g., W10), it performs poorly for others, including W06 and W18, where negative R^2 values are observed. This variability reflects the sensitivity of MPFM measurements to flow conditions, especially in low water-cut or unstable multiphase environments. Although MPFM provides reasonable MAE and RMSE values in some cases, its overall performance lacks consistency across wells.

It is important to note that MAPE is less reliable for water rate evaluation due to the presence of near-zero flow conditions. Small absolute errors can result in disproportionately large percentage errors, inflating MAPE values even when predictions are reasonably accurate in absolute terms. Therefore, MAE and RMSE provide a more meaningful assessment of model performance for water rates.

Overall, the hybrid model provides the most balanced and reliable performance for water rate prediction. While water remains the most challenging phase to model, the proposed framework indicatively improves prediction accuracy compared to both the physics-based model and the MPFM baseline, demonstrating the effectiveness of physics-informed residual learning in capturing complex multiphase flow behavior.

Table 5.3
Temporal generalization performance for water rate (q_w)

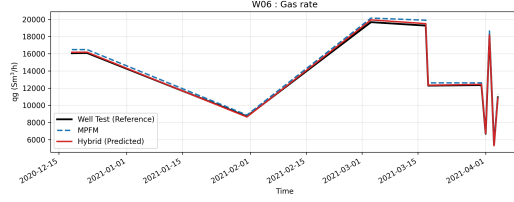
Physics Model							
Metric	W06	W08	W10	W11	W15	W18	W19
R^2	-2.46	-25.01	-0.74	-3.49	-3.12	-22.08	-15.86
MAE	2.35	15.05	19.54	21.81	5.88	52.93	20.41
RMSE	2.78	15.71	30.02	24.58	6.75	54.21	21.06
MAPE (%)	100.00	38.12	99.73	97.15	100.00	93.06	90.89
Hybrid Model							
R^2	0.97	-3.89	0.84	0.83	0.84	-0.11	0.40
MAE	0.20	6.29	5.41	4.28	0.95	11.75	3.85
RMSE	0.25	6.35	9.06	4.92	1.17	12.23	4.12
MAPE (%)	20.57	16.33	23.72	18.25	23.01	19.99	16.22
MPFM							
R^2	-0.98	0.18	0.98	0.19	0.10	-2.88	0.08
MAE	1.67	2.35	2.54	7.65	2.11	22.48	2.59
RMSE	2.06	2.60	3.10	10.76	2.78	22.84	5.09
MAPE (%)	69.96	5.90	82.23	34.02	47.85	39.62	19.06

5.2 Spatial Generalization Performance

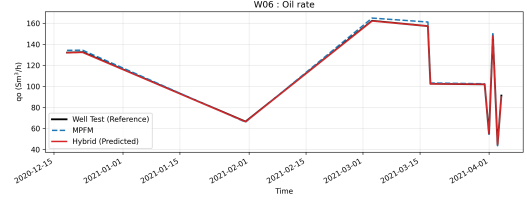
This section evaluates the ability of the proposed model to generalize across wells, i.e., to predict production rates for wells that were not observed during training. This is assessed using the Leave-One-Well-Out (LOWO) evaluation protocol described in Section 4.7, where all data from a single well is held out for testing, and the model is trained on the remaining wells. In this study, well W10 is selected as the held-out well for evaluation.

Spatial generalization represents a more challenging task than temporal generalization, as it requires the model to transfer learned relationships across wells with potentially different geological characteristics, flow regimes, and operational conditions.

(a) W06 gas rate temporal generalization performance



(b) W06 oil rate temporal generalization performance



(c) W06 water rate temporal generalization performance

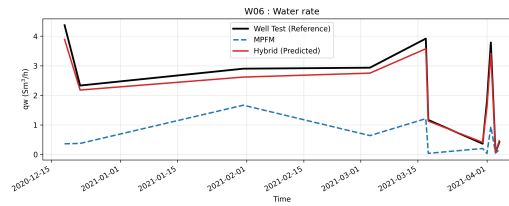
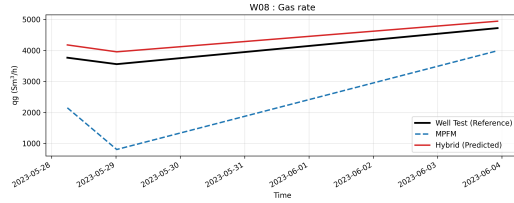


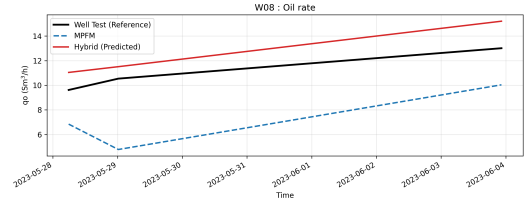
Figure 5.1

Temporal generalization performance of Physics, Hybrid, and MPFM models for W06. The plots show predicted and measured gas, oil, and water rates over time.

(a) W08 gas rate temporal generalization performance



(b) W08 oil rate temporal generalization performance



(c) W08 water rate temporal generalization performance

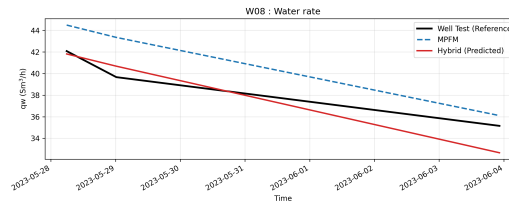
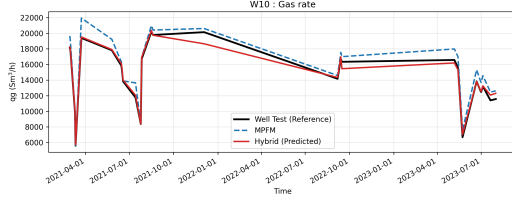


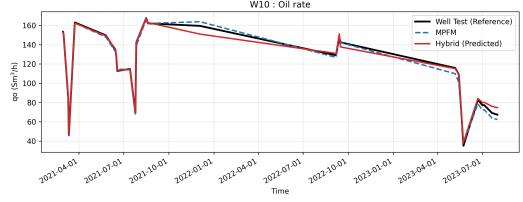
Figure 5.2

Temporal generalization performance of Physics, Hybrid, and MPFM models for W08. The plots show predicted and measured gas, oil, and water rates over time.

(a) W10 gas rate temporal generalization performance



(b) W10 oil rate temporal generalization performance



(c) W10 water rate temporal generalization performance

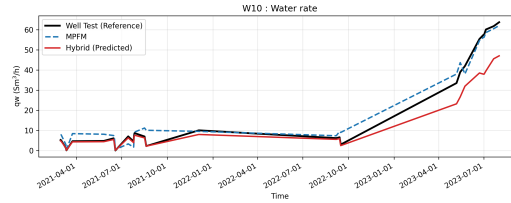
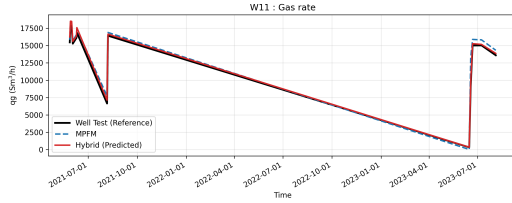


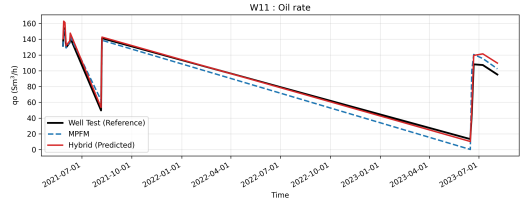
Figure 5.3

Temporal generalization performance of Physics, Hybrid, and MPFM models for W10. The plots show predicted and measured gas, oil, and water rates over time.

(a) W11 gas rate temporal generalization performance



(b) W11 oil rate temporal generalization performance



(c) W11 water rate temporal generalization performance

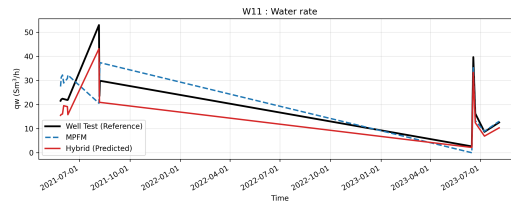
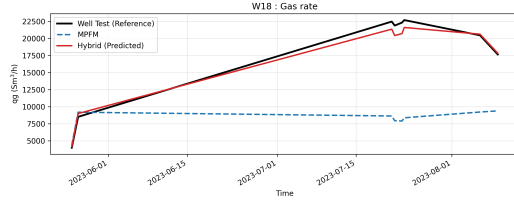


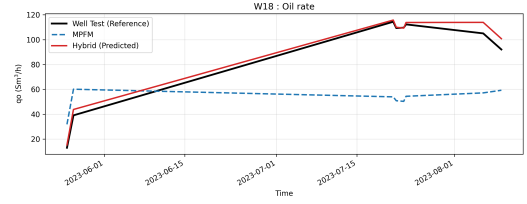
Figure 5.4

Temporal generalization performance of Physics, Hybrid, and MPFM models for W11. The plots show predicted and measured gas, oil, and water rates over time.

(a) W18 gas rate temporal generalization performance



(b) W18 oil rate temporal generalization performance



(c) W18 water rate temporal generalization performance

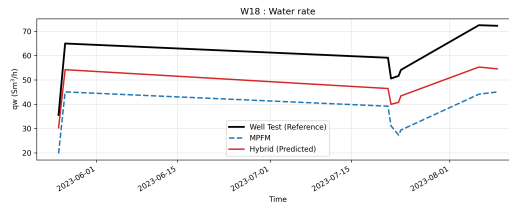
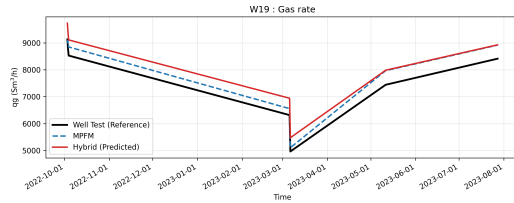


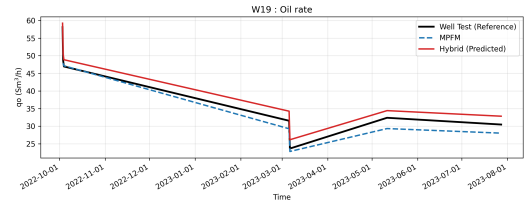
Figure 5.5

Temporal generalization performance of Physics, Hybrid, and MPFM models for W18. The plots show predicted and measured gas, oil, and water rates over time.

(a) W19 gas rate temporal generalization performance



(b) W19 oil rate temporal generalization performance



(c) W19 water rate temporal generalization performance

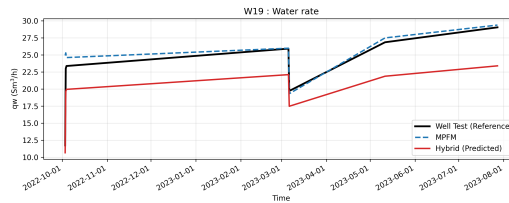


Figure 5.6

Temporal generalization performance of Physics, Hybrid, and MPFM models for W19. The plots show predicted and measured gas, oil, and water rates over time.

Therefore, strong performance under LOWO evaluation is indicative of a model’s robustness and practical applicability in real-world deployment scenarios.

5.2.1 LOWO Results

Table 5.4 summarizes the spatial generalization performance for the held-out well W10.

The physics-based model demonstrates limited generalization capability across all phases. For oil rate prediction, the model achieves a low R^2 of 0.07, indicating that it captures only a small fraction of the variance in the unseen well. Similarly, for gas rate prediction, although a moderate R^2 of 0.32 is observed, the absolute errors remain high, reflecting poor predictive accuracy. Water rate prediction is particularly poor, with a negative R^2 value, confirming the inability of the physics model to generalize across wells with different multiphase flow characteristics. These results highlight the inherent limitations of relying solely on physics-based formulations, which are often calibrated to well-specific conditions and lack adaptability to new wells.

In contrast, the hybrid model demonstrates excellent spatial generalization performance across all phases. For oil and gas rates, near-perfect accuracy is achieved, with R^2 values of 0.99 and substantial reductions in MAE and RMSE compared to both the physics model and the MPFM baseline. This indicates that the hybrid framework successfully captures transferable patterns across wells by combining physics-based predictions with data-driven residual corrections. The strong performance on the unseen well W10 suggests that the learned residual relationships are not merely overfitted to specific wells, but instead generalize effectively to new operating conditions.

For water rate prediction, the hybrid model also shows significant improvement over the physics model, achieving a high R^2 of 0.93 and low absolute errors. However, the MAPE value is notably large, which is primarily due to the presence of near-zero water rates. In such cases, small absolute errors can lead to disproportionately large percentage errors, making MAPE an unreliable metric for evaluating water rate performance. Therefore, MAE and RMSE provide a more meaningful assessment of model accuracy for this phase.

The MPFM baseline generally provides strong performance, particularly for oil and gas rates, with high R^2 values and relatively low errors. However, its performance is less consistent compared to the hybrid model, especially for water rates and under certain flow conditions. While MPFM measurements can be accurate, they are inherently dependent on sensor quality and flow regime assumptions, which may limit their reliability in complex multiphase environments.

Overall, the hybrid model achieves the best balance between accuracy and generalization capability. It significantly outperforms the physics-based model and demonstrates competitive or superior performance compared to MPFM across all phases.

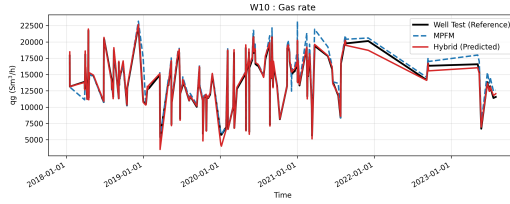
These results confirm that the integration of physics-based modeling with machine learning enables effective knowledge transfer across wells, making the proposed framework a robust solution for spatial generalization in virtual flow metering applications.

Table 5.4

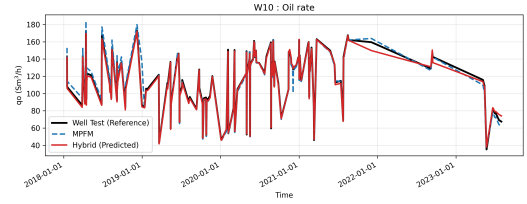
Spatial generalization (LOWO) performance for held-out well W10

Quantity	Model	R^2	MAE	RMSE	MAPE (%)
q_o	Physics	0.07	20.39	29.67	20.36
	Hybrid	0.99	1.60	2.14	1.53
	MPFM	0.97	3.07	5.55	2.82
q_g	Physics	0.32	2439.63	3050.20	19.30
	Hybrid	0.99	252.94	379.06	2.24
	MPFM	0.95	499.48	867.57	3.46
q_w	Physics	-0.25	5.39	12.15	100.00
	Hybrid	0.93	0.93	2.98	856.08
	MPFM	0.93	2.42	2.87	90809.43

(a) W10 gas rate spatial generalization performance



(b) W10 oil rate spatial generalization performance



(c) W10 water rate spatial generalization performance

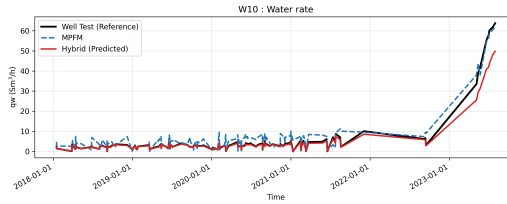


Figure 5.7

Spatial generalization performance of Physics, Hybrid, and MPFM models for W10. The plots show predicted and measured gas, oil, and water rates over time.

5.3 Performance Comparison with Contemporary Hybrid VFM

Table 5.5 presents a comparison of the proposed hybrid model against the WGR prediction models reported in (Nemoto et al., 2023).

It is important to note that while both this study and (Nemoto et al., 2023) are based on wells from the same offshore gas field, the specific wells used in each study are not necessarily identical. As a result, the comparison reflects performance across similar field conditions rather than an exact well-to-well evaluation. This distinction is important, as variability in well behavior, operating conditions, and flow regimes can influence model performance.

The proposed hybrid model demonstrates superior overall predictive performance compared to the literature models. In particular, it achieves the highest coefficient of determination ($R^2 = 0.90$), indicating a stronger ability to capture the variability in WGR across the dataset. In contrast, the models reported in (Nemoto et al., 2023) achieve lower R^2 values ranging from 0.64 to 0.80, suggesting comparatively weaker predictive capability.

In terms of error metrics, the hybrid model achieves a MAPE of 20.29%, which is comparable to the literature models (17.43%–20.61%). While one of the literature configurations reports a slightly lower MAPE (17.43%), this comes at the expense of a lower R^2 value (0.80), indicating that the model may not capture the overall variability as effectively. The proposed model, on the other hand, provides a better balance between accuracy and explanatory power.

The MRE (approximated using MPE) for the hybrid model is negative (-11.48%), indicating a tendency toward slight underprediction. In comparison, the literature models exhibit both positive and near-zero MRE values, reflecting varying degrees of bias depending on the modeling assumptions. This highlights that while the hybrid model improves overall accuracy, some systematic bias may still remain and could be further refined.

A key distinction between the proposed approach and the literature models lies in the modeling methodology. The models in (Nemoto et al., 2023) rely on different combinations of flow correlations and tuned friction factors, whereas the proposed hybrid framework integrates physics-based modeling with data-driven residual correction. The improved performance of the hybrid model suggests that learning residual corrections enables the model to capture complex deviations from simplified physical assumptions, leading to enhanced predictive capability.

Given that both studies are conducted on wells from the same field, the observed performance improvements are particularly meaningful, as they indicate that the proposed hybrid approach generalizes effectively within the same asset environment despite differences in individual well characteristics. This strengthens the validity of the comparison and highlights the robustness of the proposed framework.

Overall, the results indicate that the proposed hybrid model achieves state-of-the-art performance for WGR prediction, outperforming existing physics-based approaches reported in the literature while maintaining comparable error levels. This demonstrates the effectiveness of combining physical modeling with machine learning

for improved virtual flow metering performance.

Table 5.5

Comparison of global WGR prediction performance with contemporary hybrid VFM

Model	MRE (%)	MAPE (%)	R^2
Hybrid (Proposed)	-11.48	20.29	0.90
No-slip liquid holdup, Colebrook friction factor ¹	12.64	20.61	0.77
No-slip liquid holdup, tuned friction factor ¹	0.43	17.43	0.80
Slip liquid holdup, tuned friction factor ¹	-0.23	20.10	0.64

¹ Results reported in (Nemoto et al., 2023)

5.4 Key Observations

The following key observations can be drawn from the temporal and spatial generalization experiments:

- **Physics-only models lack generalization capability.** The physics-based model consistently underperforms in both temporal and spatial settings, with frequent negative R^2 values. This indicates that simplified physics formulations are insufficient to capture complex, evolving multiphase flow dynamics, especially under changing operating conditions and across different wells.
- **Hybrid model improves predictive accuracy.** The proposed hybrid physics machine learning model consistently achieves the lowest errors and highest R^2 values across oil, gas, and water rates. The integration of residual learning effectively corrects systematic biases in the physics model, leading to substantial performance gains.
- **Strong temporal generalization across most wells.** The hybrid model demonstrates near-perfect temporal prediction performance for oil and gas rates in multiple wells, indicating its ability to adapt to evolving production conditions over time. Performance degradation is limited to a small number of challenging wells (e.g., W08), suggesting sensitivity to underrepresented operating regimes.
- **Effective spatial generalization to unseen wells (LOWO).** Under the LOWO evaluation (with W10 as the held-out well), the hybrid model maintains high accuracy for both oil and gas rates ($R^2 \approx 0.99$), demonstrating strong transferability of learned relationships across wells. This confirms that the model captures underlying physical patterns rather than overfitting to well-specific behavior.

- **Water rate prediction remains inherently challenging.** All models exhibit relatively lower and more variable performance for water rate prediction. Although the hybrid model notably improves accuracy compared to the physics model, performance remains sensitive to sparse and intermittent water production patterns.
- **MAPE is unreliable for low-rate conditions.** Extremely high MAPE values observed for water rates are primarily due to near-zero flow conditions, where small absolute errors lead to disproportionately large percentage errors. Therefore, MAE and RMSE provide a more reliable assessment of model performance for water prediction.
- **MPFM provides a strong but inconsistent baseline.** MPFM measurements generally achieve high accuracy for oil and gas rates but exhibit instability under certain conditions (e.g., specific wells or flow regimes). This highlights limitations related to sensor reliability and flow regime assumptions in multiphase environments.
- **Hybrid model offers the best balance between accuracy and robustness.** Across both temporal and spatial evaluations, the hybrid model consistently outperforms the physics-based model and demonstrates more stable performance than MPFM. This confirms the effectiveness of combining physics-based modeling with machine learning for virtual flow metering applications.
- **Hybrid model outperforms literature physics-based approaches.** Comparison with the models reported in (Nemoto et al., 2023) shows that the proposed hybrid framework achieves higher R^2 values for WGR prediction, indicating improved ability to capture variability in multiphase flow behavior compared to traditional physics-based formulations. While some model configurations achieve slightly lower MAPE values, they do so at the cost of reduced R^2 . In contrast, the hybrid model maintains competitive error levels while achieving notably higher R^2 , demonstrating a better balance between predictive accuracy and overall model fidelity.
- **Residual learning enhances physics-based modeling.** The superior performance of the hybrid model compared to (Nemoto et al., 2023) highlights the advantage of incorporating data-driven residual correction on top of physics-based predictions. This enables the model to capture complex deviations from simplified flow correlations and friction factor assumptions used in traditional methods.
- **Consistent performance within the same field environment.** Although the wells used in this study and in (Nemoto et al., 2023) are not necessarily identical,

they originate from the same offshore gas field. The improved performance of the hybrid model under these conditions indicates its ability to generalize effectively within the same asset environment despite variations in individual well characteristics.

- **Hybrid approach demonstrates state-of-the-art performance.** Overall, the proposed hybrid model achieves performance that is competitive with or superior to existing approaches reported in (Nemoto et al., 2023), establishing it as a robust and effective solution for WGR prediction in virtual flow metering applications.

5.5 Code Availability and Reproducibility

All code implementations, experimental results, supporting documentation, and supplementary literature associated with this research are publicly available in the GitHub repository <https://github.com/pcperera/vfm>. This repository is maintained by the author and serves as the primary reference for the computational aspects of this thesis.

The repository includes the complete implementation of the proposed physics-informed residual learning framework for virtual flow metering, along with data pre-processing routines, model training and evaluation scripts, and visualization utilities used to generate the results presented in this thesis. Where applicable, documentation is provided to facilitate reproducibility and to enable further extension of the proposed methods.

CHAPTER 6

DISCUSSION

This chapter interprets the experimental findings in the context of physical modeling, data-driven learning, and practical deployment considerations.

6.1 Temporal vs Spatial Generalization

The temporal generalization results demonstrate that the hybrid model is highly effective at extrapolating future behavior within the same well when lag-safe causality constraints are enforced. Across multiple wells, the hybrid model achieves near industry-standard accuracy for oil and gas rates, significantly outperforming the physics-based model and consistently matching or exceeding the performance of MPFM, which serves as an industry-standard benchmark. This indicates that the residual errors learned by the model are largely stable over time within a given well, enabling consistent correction of systematic physics bias under evolving operating conditions.

In contrast, spatial generalization is inherently more challenging, as it requires the model to transfer learned relationships across wells with different characteristics. Despite this, the hybrid model demonstrates strong transferability under the LOWO evaluation (with W10 as the held-out well), achieving near industry-standard accuracy for oil and gas rates. This suggests that the learned residual corrections capture underlying physical deviations that are consistent across wells, rather than being purely well-specific. The ability to generalize to an unseen well further confirms that the model is not overfitting to individual well behavior, but instead learning transferable representations of multiphase flow dynamics.

However, some performance variability is still observed across wells in the temporal setting (e.g., W08), highlighting the sensitivity of the model to underrepresented operating regimes. This indicates that while the hybrid model generalizes well, its performance is influenced by the diversity and coverage of training data. Compared to MPFM, the hybrid model provides more consistent performance across wells, whereas MPFM exhibits instability under certain flow conditions, reflecting its dependence on sensor assumptions and flow regime behavior.

6.2 Comparison with Contemporary Hybrid VFM

The comparison with the models reported in (Nemoto et al., 2023) provides additional context for evaluating the effectiveness of the proposed approach. While both studies are based on wells from the same offshore gas field, the specific wells used are not

necessarily identical. Therefore, the comparison reflects performance under similar field conditions rather than a direct well-to-well evaluation.

Despite this, the proposed hybrid model demonstrates superior performance in terms of explanatory power, achieving higher R^2 values compared to the physics-based configurations presented in the (Nemoto et al., 2023) model. Although some literature configurations achieve slightly lower MAPE values, these are typically associated with lower R^2 , indicating a reduced ability to capture variability in WGR behavior.

The improved performance of the hybrid model can be attributed to its ability to learn residual corrections on top of physics-based predictions. In contrast, the models in (Nemoto et al., 2023) rely on variations of flow correlations and tuned friction factors, which may not fully capture complex deviations from idealized physical assumptions. The results therefore suggest that augmenting physics-based models with data-driven residual learning provides a more flexible and accurate framework for virtual flow metering.

Importantly, the fact that both studies are conducted within the same field environment strengthens the validity of this comparison. The observed improvements indicate that the hybrid approach generalizes effectively within the same asset, even when applied to different wells with varying operational characteristics.

6.3 Deployment Implications

The results demonstrate that the hybrid framework offers a practical and scalable alternative to traditional MPFM systems, particularly in environments where dense instrumentation is unavailable or cost-prohibitive. While MPFM provides strong performance in many cases, it should be noted that it is not a true ground-truth measurement but rather an industry-standard benchmark subject to measurement uncertainty and flow regime limitations. As observed in the experiments, MPFM performance can degrade under complex multiphase conditions.

The hybrid model, on the other hand, achieves substantial improvements over physics-only approaches while maintaining competitive performance relative to MPFM. Its ability to generalize both temporally and spatially makes it well-suited for deployment in brownfield assets, where historical data is available but measurement infrastructure may be limited.

Furthermore, the use of lag-safe temporal splitting ensures that the evaluation reflects realistic deployment scenarios, where future data is not accessible during model training. This strengthens the validity of the reported results and demonstrates that the model can operate under causal constraints.

An additional advantage of the hybrid approach is its ability to leverage existing physics knowledge while adapting to data-driven corrections, reducing reliance on extensive sensor calibration and manual tuning required in traditional systems.

6.4 Summary

In summary, the experimental results confirm that:

- Hybrid physics-ML modeling indicatively improves production rate estimation under both temporal and spatial generalization, particularly for oil and gas rates.
- The hybrid model demonstrates strong transferability across wells, achieving near-perfect performance under LOWO evaluation (W10) for unseen wells.
- Comparison with the (Nemoto et al., 2023) model shows that the proposed approach achieves superior explanatory power while maintaining competitive error levels.
- Lag-safe data partitioning is essential for ensuring causal and scientifically valid evaluation of time-series models.
- Water rate prediction remains challenging due to sparse and intermittent flow behavior, and percentage-based metrics such as MAPE can be misleading in low-rate conditions.
- MPFM serves as a strong industry benchmark but exhibits variability under certain flow conditions, highlighting the need for robust data-driven alternatives.
- The hybrid approach provides a robust and practical alternative to both physics-only models and MPFM, offering improved consistency across varying operating conditions.

These findings support the central thesis that physics-informed residual learning can bridge the gap between deterministic models and fully data-driven approaches, enabling accurate, robust, and deployable solutions for real-world well production monitoring.

CHAPTER 7

FUTURE WORK

The physics-informed residual learning framework developed in this study demonstrates promising performance for virtual flow metering under sparse and irregular measurement conditions. While the proposed approach addresses several practical limitations of conventional flow metering techniques, there remain multiple avenues for further research and enhancement. This chapter outlines potential directions for future work aimed at improving model fidelity, robustness, scalability, and industrial applicability.

7.1 Extension of Physics-Based Modeling

The current physics layer employs a simplified steady-state formulation to model liquid and gas production based on pressure, temperature, and choke measurements. Future work could extend this formulation to incorporate more detailed multiphase flow physics, including frictional pressure losses, phase slip effects, and temperature-dependent fluid properties. Incorporating transient flow equations or quasi-steady corrections may further improve performance under highly unsteady operating conditions, such as during choke adjustments or flow instabilities.

In addition, the physics model could be enhanced by integrating well-specific geological and completion parameters, such as tubing roughness, completion depth, or reservoir connectivity, thereby reducing the reliance on purely data-driven corrections.

7.2 Advanced Residual Learning Architectures

The residual learning component in the current framework is implemented using tree-based ensemble models for computational efficiency and robustness. Future research could investigate alternative residual learning architectures, including recurrent neural networks or attention-based temporal models, to better capture time-dependent error structures. Such models may be particularly beneficial for wells exhibiting strong temporal correlations or delayed flow responses.

Hybrid residual architectures that combine lightweight temporal models with regime-aware gating mechanisms could further improve accuracy while maintaining computational tractability.

7.3 Dynamic Regime Detection and Adaptive Gating

In the present framework, residual corrections are selectively applied using predefined gating rules based on physical heuristics such as water cut thresholds and residual magnitude. Future work could focus on developing adaptive or data-driven regime detection mechanisms that automatically identify flow regimes and dynamically adjust correction strategies. Probabilistic gating or soft-switching approaches may provide smoother transitions between physics-dominated and data-driven predictions.

7.4 Uncertainty Quantification and Confidence Estimation

The current model produces point estimates of flow rates without explicit uncertainty bounds. Future research could incorporate uncertainty quantification techniques, such as Bayesian residual learning or ensemble-based confidence estimation, to provide probabilistic flow predictions. Quantifying predictive uncertainty is critical for operational decision-making, production forecasting, and risk assessment in field-scale applications.

7.5 Validation Across Diverse Field Conditions

Future studies should validate the proposed framework across a wider range of reservoirs, well types, and production conditions. Applying the model to different asset classes, including gas-dominated wells, mature waterfloods, and unconventional reservoirs, would further assess its generalizability and robustness. Such validation would strengthen confidence in the framework's applicability beyond the specific datasets considered in this study.

Overall, these future research directions aim to enhance the physical realism, adaptability, and industrial relevance of physics-informed residual learning approaches for virtual flow metering, building upon the foundational architecture developed in this work.

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